

OEIS RESEARCH REPORT

Fixed-composition excursions

D-finiteness, transcendence, and all-order
asymptotics for OEIS A215561

With an explicit correction for A215570
and asymptotic inversion

Merged from two independent manuscripts of the same day:

Fixed-composition excursions and *Balanced Late-Growing Permutations*; Part IV from a third

Prepared for Vladimir Reshetnikov

October 1, 2026; Part IV added October 2, 2026

Abstract

Let $A_r(n)$ count words containing n copies of each letter in $\{1, \dots, r\}$ whose every prefix has sum at most its average over all such words. We prove the fixed-row asymptotic conjecture recorded in OEIS A215561, identify its constant as a positive algebraic number, and establish a complete expansion in inverse powers of n . The central formula is

$$A_r(n) \sim \frac{\kappa_r}{rn} \frac{(rn)!}{(n!)^r}, \quad \kappa_r = \exp \left(\sum_{m \geq 1} \frac{\mathbb{P}(S_m = 0)}{m} \right),$$

where S_m is the centered, uniformly weighted walk associated with the alphabet. A finite kernel-root product computes κ_r .

We also prove that every fixed row is D-finite, while its ordinary generating function is transcendental for every $r \geq 3$. This separates recurrence existence from algebraicity and from the specific recurrence conjecture for A215570 (proved in Part IV). An explicit first correction for that sequence, a fixed-composition contact law, and an all-order Lambert- W inversion procedure are derived. Exact enumeration and symbolic checks accompany the proofs.

Part II adds the second manuscript's independent results: an elementary one-large-summand proof of the leading term for every finite labelled step set and rational balanced composition, a second all-orders route with a Gaussian-moment algorithm, second-order terms for $r = 2, 3$, a single giant primitive component with independent finite decorations and a heavy-tailed gap, eventual strict log-convexity, and an explicit rounding bound.

Part IV, from a third manuscript, proves the order-three recurrence that Kauers and Koutschan conjectured for A215570, by a positive Legendre-coefficient formula and a rational telescoping certificate, and computes $\alpha_{5,2}, \dots, \alpha_{5,8}$ exactly.

Status. This is a newly written mathematical research manuscript, not a peer-reviewed publication or a Lean-verified development. Classical kernel and diagonal theorems are credited separately from the applications and deductions proved here. No claim of exhaustive bibliographic priority is made. Parts I and II do *not* prove the particular order-three recurrence conjecture of Kauers and Koutschan; Part IV (added October 2) does. The second manuscript describes itself as an “AI-assisted mathematical research draft. Unrefereed; not formally verified.”

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1 Sources, merge decisions and conventions

1.1 Two independent manuscripts

This report merges two manuscripts of batch 73 of the repository’s incoming-reports intake (cluster O2). Both are dated October 1, 2026, both were prepared for Vladimir Reshetnikov, and both were placed in commit 6e193dd4f.

- **Source 61** (batch-73 manuscript 61): *Fixed-composition excursions. D-finiteness, transcendence, and all-order asymptotics for OEIS A215561. With an explicit correction for A215570 and asymptotic inversion* (a 1,505-line source and a 25-page PDF). Archive `oeis_balanced_excursions.zip`, arrival commit `aa43cc555`; it pins the repository at `40b17f2fc`.
- **Source 42** (batch-73 manuscript 42): *Balanced Late-Growing Permutations. Algebraic constants, D-finiteness, and complete asymptotic expansions. A proof of the fixed-row growth conjecture for OEIS A215561* (a 953-line source and a 22-page PDF, with the banner “AI-assisted mathematical research draft. Unrefereed; not formally verified”). Archive `late_growing_excursions.zip`, arrival commit `bdf1a1a73`; it pins the repository at `bc4d1fa2b`.

The two were written independently on the same day and share no prose: about 1.8% of their 8-word shingles are common in each direction, all of them bibliography entries, OEIS rows and kernel polynomials. Source 42’s pin predates both arrivals. Source 61’s pin is later than source 42’s arrival and contains source 42’s archive as an unopened zip file, so source 61’s repository search (Section 2.3) could not have found it. Their 47 common exactly computed terms are equal, and their constants agree to the last printed digit for $r \leq 8$. Source 61 is the base text; Part II holds what only source 42 has.

(Addition of October 2, 2026.) A third independent manuscript on the same problem arrived in batch 75 (manuscript 03, *Fixed-Content Ballot Words*, here **source 75-03**; archive `OEIS_Fixed_Content_Ballot_Words.zip`, arrival commit `4b874cea0`, placed in `6ea60e367`, pin `29aca108e`). Part IV prints what it adds, chiefly a proof of the Kauers–Koutschan recurrence for A215570; Section 24 gives its provenance and Table 5 the results it re-proves, which are not printed again.

1.2 What each source contributed

Table 1 lists every result and where it is printed. “Both” means the two sources prove the result independently; it is then printed once, with source 61’s statement, and source 42’s proof is printed as well when it follows a different route.

1.3 Where the merge chose

Source 61 is the base: it carries the larger set of theorems about the OEIS family (transcendence, the field $\mathbb{Q}(\sqrt{5})$, all orders and D-finiteness on rational rays, the Hessian determinant, the integral formula, the $r = 5$ inverse numerics). Source 42’s leading-term theorem has weaker analytic hypotheses (finite bridge mass, no saddle point) and a wider class of step sets, so the usual preference for the more general source would make it the base for that theorem alone; it is printed instead as the general theorem of Part II, with second-route status for the rows, because source 61 is stronger on its rays.

Table 1: Results of the sources. The last three rows are source 75-03's additions (Part IV); its re-proofs of other rows are listed in Table 5.

result	sources	printed as
fixed-row asymptotic, constant κ_r , all orders	both	Theorem 3.1; second routes Theorem 16.1, Section 17
D-finiteness of every row	both	Theorem 3.2
transcendence of every row with $r \geq 3$	61	Section 8
kernel product and bridge identity (classical)	both	Lemmas 4.1, 5.1, with second proof
root-product formula for κ_r	both	Proposition 5.2
integral formula for $\log \kappa_r$	61	(5.7)
sextic for $r = 5$	both	(4.5), Remark 4.2
Hessian determinant, normalization check	61	Lemma 6.1
$\alpha_{5,1} = -13/10 + 13\sqrt{5}/50$	both	Theorem 3.3; Sections 9, 18
$\alpha_{5,j} \in \mathbb{Q}(\sqrt{5})$ for all j	61	Theorem 3.3
second-order terms for $r = 2, 3$; $\alpha_{3,1}$ from the general formula	42	(18.6)
law of the number of arches	both	Theorem 10.1; Theorem 19.2 (general compositions)
giant primitive component, decoration tail	42	Section 19
eventual strict log-convexity	42	(20.2)
leading term for any finite labelled step set	42	Theorem 16.1
all orders and D-finiteness on centred rational rays	61	Corollary 12.1
Lambert- W inverse	both	Section 11
explicit rounding bound	42	(20.4)
constants: $r \leq 8$ (61), $r \leq 12$ (42)	both	Table 3
exact enumerations	both	Sections 13, 21
research directions: eight (61), ten (42)	both	Section 22
Kauers–Koutschan Conjecture 15: the A215570 recurrence	75-03	Theorem 26.1, Appendix C
positive Legendre formula for A215570	75-03	Theorem 25.2
exact $\alpha_{5,2}, \dots, \alpha_{5,8}$; second-order inverse for $r = 5$	75-03	Proposition 27.1, (28.1)

Source 42's proofs are kept where they follow a genuinely different route: the convolution proof of the leading term (Theorem 16.1); the factorization proof of the bridge identity (after Lemma 5.1); the all-orders proof in lattice coordinates with its coefficient algorithm (Section 17); the first-correction formula (Section 18); the elimination for κ_4 (Section 7); the cutoff proof of the arch law (Theorem 19.2); and the second way of removing the auxiliary variable in the kernel lemma. Not printed a second time: source 42's statements of the main theorem, of D-finiteness and of the root-product constant, its sextic (the negative of (4.5)), its kernel lemma and D-finiteness proof, and its Lambert- W core with first corrections, all of which coincide with source 61's. The two constant tables are merged, the research directions are merged item by item, and source 61's package appendix is rewritten for the shipped file names, with source 42's instructions folded into it.

For source 75-03 (Part IV) the merge prints in full only what Parts I and II lack: the positive coefficient formula, the proof of the recurrence with its certificate (Appendix C), the coefficient algorithm and the exact $\alpha_{5,j}$. Its two-variable saddle is kept, briefly, as a second route to the $r = 5$ expansion, because it supplies $\alpha_{5,1}$ by a third contraction, and its resolvent derivation of the sextic as a remark. Everything else it proves is in Parts I and II already, mostly in a stronger form, and appears only as a row of Table 5, with two small increments stated in

Section 24.2. Its research questions are merged into Section 22; its package description is folded into Appendix B.

1.4 Repository context and re-derived material

The bridge-excursion identity (Lemma 5.1 and (14.3)) is the composition-marked case, with coefficient ring $\mathbb{Q}[x_1, \dots, x_r]$ and step symbol $\sum_i x_i u^{s_i}$, of Corollary `cor:excursion` of the collection’s report *Multiple-chain preorder polytopes: two proofs of the exponential formula* (`SetTheory/Cardinals/docs/reports/enumerative-combinatorics/multiple-chain-exponential-formula`), proved there over any commutative coefficient ring from its meander factorization; it is also classical [8]. Both sources’ proofs are kept as further routes.

The inversion of Section 11 and the rounding bound (20.4) re-derive, for this family, the apparatus of the chapter “Apparatus for inverting a rapidly growing function” (`p0:sec:top`) of the canonical transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`: its exact-core and perturbed inversion (`p0:thm:perturbed-inversion`) and its staircase theorem `p0:thm:staircase` for integer thresholds. The separation clause of the staircase theorem and its failure for arbitrarily small error are formalized as the Lean theorems `Fabius.staircase_separation` and `Fabius.staircase_separation_fails` in `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`. Neither source cites these theorems (source 61 cites the README of that volume, unchanged since its pin [11]); they are cited here, and no novelty is claimed for the inversion or rounding statements. The same holds for source 75-03’s inversion of A215570, which also does not cite them (Section 28).

All three sources are AI-assisted and unrefereed. No Lean or Rocq development checks any statement of this report; the Lean theorems just named are general facts about ceilings and do not verify its asymptotic inputs, and its place in the research-report collection confers no formal status.

1.5 Notation

The two sources used different letters for the same objects and the same letters for different objects. This report fixes: $A_r(n)$ for the counts; κ_r for the constant and $\kappa(\mathbf{p})$ for a composition $\mathbf{p}$; $\alpha_{r,j}$ for the normalized coefficients of the expansion; b_j for the coefficients of its logarithm and μ_j for the coefficients of the inverse series; $N_r(Y)$ for the integer threshold (a bare N is always a length); $B(t; \mathbf{x})$ for the bridge generating function and $\mathcal{L}(\mathbf{x}) = \log \mathcal{E}(\mathbf{x})$ for the log-excursion (or log-bridge) series, whose marked specialization is the $\mathcal{L}(t; \mathbf{y})$ of Section 6 (the calligraphic $\mathcal{B}$ of (9.2) is a length correction, unrelated); H for the Hessian in marker coordinates and Q for the corresponding matrix in the lattice coordinates of Part II (not the kernel polynomial $Q_r(u)$); ν for the time period; $\mathcal{I}$ for the generating function of arches (primitive excursions); h for the depth of the most negative step; and V for the step variance. Table 2 lists every renamed symbol; no normalization changed (source 75-03’s Q_r has the opposite sign; its other conventions are fixed in Section 24.3). A few letters keep two meanings in separate places, where the context fixes them: q is the number of labelled steps in Part II, the tilted probabilities $q_i(\mathbf{y})$ in Section 6, and a root product in (4.5); X is a Gaussian vector in Part II and a discrete random vector in the proof of Lemma 6.1; Y is a target value in the inversions and Y_i a random step in Lemma 15.1; and the bridge weights b_k carry a composition index, the logarithmic coefficients b_j a scalar one.

Table 2: Renamed symbols.

source 42	here	source 42	here
$T_r(n)$	$A_r(n)$	C, c_i (projection)	Ξ, ξ_i
$E_r, E(\mathbf{p}), E$	$\kappa_r, \kappa(\mathbf{p}), \kappa$	V (lattice basis)	$\mathbf{V}$
$B_r = \log E_r$	$\log \kappa_r$	$R(z)$	$P_*(z)$
B (bridge mass)	$\mathfrak{B}$	$\lambda(z), \lambda_m$	$\Psi(z), \Psi_m$
$\mathcal{B}$ (log-bridge)	$\mathcal{L}$	$E(t, z), E_*(z)$	$\widehat{\mathcal{E}}(t, z), \kappa_*(z)$
$d_j(r)$	$\alpha_{r,j}$	H_{an}, H_k	$\Upsilon_{\text{an}}, \Upsilon_k$
$q_j(r)$	b_j	G_j	$\mathcal{G}_j$
C_r	K_r	β, t_h (transfer)	γ, t_i
α_r, λ_r	β, λ	β_j	χ_j
$n_r(Y)$	$N_r(Y)$	$h = N^{-1/2}$	$\hbar$
$K_{r,J}$, truncation J	$\varpi_{r,M}, M$	η (exponent)	ε_1
δ (period), σ^2	ν, V	M_j, M_{12}	$\mathcal{M}_j, \mathcal{M}_{12}$
a (minimum step)	h	$a(z)$	$\mathbf{a}(z)$
$\alpha = (q+1)/2, a_0$	ς, η	ρ (nonprincipal roots)	ω
K_N (primitive factors)	J_N	$v_1, v_2, v_3; s$	$a_1, a_2, a_3; w$
J_N (giant component)	Ω_N	$h(\mathbf{k})$ (height)	$\text{ht}(\mathbf{k})$
U_N, V_N, U, V	$\mathcal{U}_N, \mathcal{V}_N, \mathcal{U}, \mathcal{V}$	$D(t)$	$\Theta(t)$
D (decoration length)	D_{dec}	U_j, V_j (roots)	$u_j, \tilde{u}_j$
$D(\ \cdot\)$	$D_{\text{KL}}(\ \cdot\)$	Y (sextic)	X
source 61	here	source 61	here
d_j (inverse series)	μ_j	H_0 (arches)	$\mathcal{I}$
$T_r(Y)$	$N_r(Y)$		
source 75-03	here	source 75-03	here
$\mathcal{E}_r, \mathcal{E}(\mathbf{p})$	$\kappa_r, \kappa(\mathbf{p})$	d_j (asymptotic)	$\alpha_{5,j}$
$E(\mathbf{x}), e(\mathbf{k})$	$\mathcal{E}, e_{\mathbf{k}}$	$H(\mathbf{x}), H_0$	$\mathcal{L}, \mathfrak{B}$
$h(\mathbf{k}), D = \frac{r+1}{2}$	$b_{\mathbf{k}}, \varsigma$	M (cutoff), ℓ (summands)	H, m
$K_N; q = 1 - \mathcal{E}^{-1}$	$J_N; \eta$	$Q_r(u)$	$-Q_r(u)$
z (length); b (depth)	$t; h$	$G_r(z); B(z)$	$F_r(z); B(t; \mathbf{x})$
Y (sextic); A, B, C, T	$X; \mathbf{A}, \mathbf{B}, \mathbf{C}, \bar{c}$	d_n (zero-free count)	$A_5^{\circ}(n)$
C_n (Legendre coefficient)	Λ_n	P_n (Legendre)	$\mathcal{P}_n$
$W; N, M; p, q, k$ (Lagrange)	$\mathcal{Y}; \mathbf{n}, \mathbf{n}+1; i, j, k$	$\tilde{a}_n, h_n$	$\tilde{A}_5(n), \mathfrak{h}_n$
R, S, U, V	$\mathfrak{R}, \mathfrak{S}, \mathfrak{U}, \mathfrak{V}$	$p_j, q_j, \lambda(n)$	$\mathfrak{p}_j, \mathfrak{q}_j, \mathfrak{c}(n)$
w_n, L_n	$\mathcal{W}_n, \mathcal{L}_n$	$V_j, D = t^2 - 1$	$\zeta_j, t^2 - 1$
$\mathcal{S}_0, \mathcal{S}_1; \mathcal{A}_n, \mathcal{B}_n$	$\mathcal{R}_0, \mathcal{R}_1; \mathcal{U}_n, \mathcal{V}_n$	$\alpha_j, \beta_j; R_m, S_m$	$\mathbf{u}_j, \mathbf{v}_j; \mathfrak{R}_{\sharp}, \mathfrak{S}_{\sharp}$
$\mathcal{P}(u, v); H$	$\Pi(u, v); H_{\Pi}$	$\rho; G = 1 + u_0$	$\varphi^{-2}; G_0$
$\eta_j; c_k; P_j(z)$	$\eta_j^{\Lambda}; \tilde{c}_k; \widehat{\mathfrak{p}}_j(z)$	$b_j, g_j; \kappa_j; (X, Y)$	$\ell_j^+, \ell_j^-; \mathcal{K}_j; (Z_1, Z_2)$
$L = 5 \log 5; y; N(y)$	$\lambda; Y; N_5(Y)$	D (log variable)	D_Y

Part I

Fixed-composition excursions

2 The problem, its significance, and the source audit

2.1 A natural multiset-permutation family

For a fixed integer $r \geq 2$, let $A_r(n)$ be the number of words $w = w_1 \cdots w_{rn}$ containing exactly n occurrences of each letter $1, \dots, r$ and satisfying

$$\sum_{j=1}^k w_j \leq \frac{r+1}{2}k \quad (0 \leq k \leq rn). \quad (2.1)$$

These are rows of the array A215561. The average on the right is exact: in a uniformly chosen multiset permutation, each position has expected letter $(r+1)/2$. This family includes Catalan numbers and a balanced Motzkin-word enumeration, but already the fourth and fifth rows lead to substantially more difficult coefficient extraction. The first rows are balanced versions of finite-step excursion enumeration, combining a ballot constraint with prescribed letter multiplicities; fixing the multiplicities is the significant extra condition, and the unrestricted-composition excursion sequence and a fixed-composition diagonal are different objects (source 42). Source 42 calls the counted words *late-growing permutations*, after the titles of the OEIS entries, and imposes (2.1) for $1 \leq k \leq rn$, which is equivalent.

The relevant sequence identifiers are

r	OEIS entry	Description of the row
2	A000108	Catalan numbers
3	A007004	Balanced three-letter excursions
4	A215562	Four-letter prefix-average words
5	A215570	Five-letter prefix-average words
6	A215571	Six-letter prefix-average words
7	A215593	Seven-letter prefix-average words

The array entry records Kotesovec's 2016 conjecture that, for each fixed $r > 1$,

$$A_r(n) \sim c(r) \frac{r^{rn}}{\pi^{(r-1)/2} n^{(r+1)/2}}. \quad (2.2)$$

The fourth and fifth entries already supply explicit leading constants; those constants are credited to Kotesovec, rather than presented here as new discoveries [1, 2, 3].

2.2 What is and is not resolved

Kauers and Koutschan discuss A215570 and A215562 in Section 6.1 of their 2023 paper. Their finite transfer matrices grow with n , which prevents their argument from proving D-finiteness. They leave a particular order-three recurrence for A215570 conjectural and ask for a provably correct recurrence for A215562 [7]. We replace the growing matrix by a fixed algebraic generating function in composition-marking variables. (2026-10-02 note: Part IV proves that order-three recurrence, Theorem 26.1; the A215562 request remains open.)

The main results below establish the existence of polynomial-coefficient recurrences for *every* fixed row. They do not print a recurrence for the fourth row and do not certify the particular guessed operator for the fifth (Part IV now certifies it). Recurrence existence, an explicit annihilating operator, and minimality of that operator are three different assertions. Source 42 draws the same distinctions and divides the credit the same way: the kernel method and the bridge–excursion identity are classical ([8], Theorem 2 and Corollary 2), closure of D-finite series under diagonals is classical [9], and the contribution is their use with a directional coefficient theorem to resolve the fixed-composition growth assertion, determine its constants, and obtain the additional fixed-composition conclusions.

The source check used the live OEIS entries, the cited 2023 paper, the classical directed-path kernel theory, and targeted searches for the sequence identifiers. The fixed-row asymptotic was still labelled a conjecture in A215561 when checked. No proof of the precise results below was identified in that search. This is a bounded source audit, not a proof that no such result exists anywhere in the literature.

2.3 Relationship to ProveIt

The inspected ProveIt snapshot was

40b17f2fc79a10a6fc1a50ca9c0c3d6f77be69e5.

Its transseries-and-inversion documentation distinguishes exact cores, finite-order expansions, remainder transport, and discrete staircase inversion. We follow that organization in Section 11; the theorems of that volume which the inversion re-derives are named in Section 1.4. The current path inspected was

Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/README.md.

The document’s own formalization inventory is deliberately narrower than its mathematical exposition [11]. No repository theorem is silently imported as a formal proof of our excursion results. Searches for A215561 and A215562 returned no repository matches; that limited check is not an audit of every file or of other identifiers for the same problem. (2026-10-01 placement note: manuscript 42, *Balanced Late-Growing Permutations*, delivered in bdf1a1a73 and present at the pin as a zipped archive, proves the same leading theorem independently; the two are merged in this report, Section 1.) Source 42 inspected the repository at its own pin bc4d1fa2b [12]; its targeted search did not locate an A215570 report, which was true then, and it states that neither that search nor its literature search is an exhaustive novelty certificate, and that it is mathematically independent of unverified repository theorems.

3 Main theorems

Put

$$g_r = \gcd(2, r-1), \quad s_i = \frac{2i - r - 1}{g_r}, \quad h = -s_1, \quad P_r(u) = \frac{1}{r} \sum_{i=1}^r u^{s_i}. \quad (3.1)$$

For odd $r = 2h + 1$, the steps are $-h, -h + 1, \dots, h$. For even r , they are the odd integers $-(r-1), -(r-3), \dots, r-1$. Let S_m be the sum of m independent steps chosen uniformly from this set.

Theorem 3.1 (All fixed rows: leading term and complete expansion). *For every fixed $r \geq 2$, the series*

$$\log \kappa_r = \sum_{m \geq 1} \frac{\mathbb{P}(S_m = 0)}{m} \quad (3.2)$$

converges, and κ_r is a positive real algebraic number. For each integer $M \geq 0$ there are real algebraic numbers $\alpha_{r,1}, \dots, \alpha_{r,M}$ such that

$$A_r(n) = K_r \frac{r^{rn}}{n^{(r+1)/2}} \left(1 + \sum_{j=1}^M \frac{\alpha_{r,j}}{n^j} + O_{r,M}(n^{-M-1}) \right), \quad (3.3)$$

where

$$K_r = \frac{\kappa_r}{\sqrt{r} (2\pi)^{(r-1)/2}}, \quad c(r) = \frac{\kappa_r}{\sqrt{r} 2^{(r-1)/2}}. \quad (3.4)$$

In particular,

$$\frac{A_r(n)}{(rn)!/(n!)^r} \sim \frac{\kappa_r}{rn}. \quad (3.5)$$

All coefficients in (3.3) are computable by finite algebraic operations on kernel expansions and Gaussian moments.

The constants implied by the O -notation are allowed to depend on r and M . Nothing in this statement is uniform when the alphabet size grows with n .

Theorem 3.2 (Algebraic ambient series; transcendental row series). *The multivariate generating function counting nonnegative excursions with steps (3.1), marking every step type independently, is algebraic over $\mathbb{Q}(x_1, \dots, x_r)$. Consequently*

$$F_r(z) = \sum_{n \geq 0} A_r(n) z^n$$

is D-finite for every fixed r . The function F_r is transcendental over $\mathbb{Q}(z)$ for every $r \geq 3$. For $r = 2$ it is the algebraic Catalan generating function.

Thus taking a diagonal preserves D-finiteness here, but does not preserve algebraicity. The transcendence assertion uses not just the growth exponent but, for even r , the power of π in the leading constant.

Theorem 3.3 (An explicit next term for A215570). *For the fifth row,*

$$A_5(n) = \frac{3\sqrt{5} - 5}{8\pi^2} \frac{5^{5n}}{n^3} \left(1 - \frac{13(5 - \sqrt{5})}{50n} + O(n^{-2}) \right). \quad (3.6)$$

Moreover every coefficient $\alpha_{5,j}$ in its full expansion belongs to $\mathbb{Q}(\sqrt{5})$.

The leading factor in (3.6) is the previously posted OEIS factor. The proof and the correction are developed below without assuming the conjectured recurrence.

Sources. Theorem 3.1 and the D-finiteness part of Theorem 3.2 are proved in both sources. Source 42 states them as its Theorems 1.1 and 1.2, with $1 < \kappa_r < \infty$; it proves the leading term without a saddle point (Theorem 16.1) and the complete expansion in lattice coordinates (Section 17). The transcendence part of Theorem 3.2 and the field statement of Theorem 3.3 are source 61's alone; the value of $\alpha_{5,1}$ was computed in both, by different contractions (Sections 9 and 18).

4 The exact excursion generating function

4.1 Encoding the prefix condition

Replacing a letter i by $r + 1 - i$ reverses the prefix inequality and preserves the multiset. It is therefore equivalent to count words whose centered partial sums, using steps s_i , are *nonnegative*. The total centered sum is zero. These are one-dimensional excursions with prescribed numbers of the individual steps.

For a count vector $\mathbf{k} = (k_1, \dots, k_r)$ write $|\mathbf{k}| = \sum_i k_i$ and $\mathbf{x}^{\mathbf{k}} = \prod_i x_i^{k_i}$. Let $e_{\mathbf{k}}$ count nonnegative excursions with that composition, and set

$$\mathcal{E}(\mathbf{x}) = \sum_{\mathbf{k} \geq 0} e_{\mathbf{k}} \mathbf{x}^{\mathbf{k}}. \quad (4.1)$$

The empty excursion contributes one. A nonzero coefficient requires $\sum_i s_i k_i = 0$, and

$$F_r(z) = \text{Diag } \mathcal{E}(z) := \sum_{n \geq 0} [x_1^n \cdots x_r^n] \mathcal{E}(\mathbf{x}) z^n. \quad (4.2)$$

In particular, the dimension of this ambient generating function is fixed when r is fixed.

4.2 The small-root formula, with a proof

Temporarily mark total length by t , and write

$$P_{\mathbf{x}}(u) = \sum_{i=1}^r x_i u^{s_i}.$$

The kernel equation is

$$u^h (1 - t P_{\mathbf{x}}(u)) = 0. \quad (4.3)$$

Of its $2h$ roots, exactly h tend to zero as $t \rightarrow 0$; these are called small roots. This description may be understood in a Puiseux extension with generic step weights.

Lemma 4.1 (Kernel product). *If $u_1(t), \dots, u_h(t)$ are the small roots, then*

$$\mathcal{E}(t\mathbf{x}) = \frac{(-1)^{h+1}}{tx_1} \prod_{j=1}^h u_j(t). \quad (4.4)$$

Consequently $\mathcal{E}$ is an algebraic multivariate formal power series.

Proof. Let $G_j(t)$ count nonnegative paths ending at height j , and put $G(t, u) = \sum_{j \geq 0} G_j(t) u^j$. Appending one step gives

$$(1 - t P_{\mathbf{x}}(u)) G(t, u) = 1 - t \sum_{s_i < 0} x_i \sum_{j=0}^{-s_i-1} G_j(t) u^{j+s_i}.$$

After multiplication by u^h , the right-hand side is a monic polynomial $N(t, u)$ of degree h . Substitution of any small root is legitimate in the formal path equation and gives $N(t, u_j(t)) = 0$. Hence $N(t, u) = \prod_{j=1}^h (u - u_j(t))$. Its constant coefficient is $-tx_1 G_0(t)$: only the most negative step, followed from height zero, contributes to that coefficient. Equating constant coefficients proves (4.4), since $G_0(t) = \mathcal{E}(t\mathbf{x})$.

The roots are algebraic over $\mathbb{Q}(t, \mathbf{x})$, so their product is algebraic. An algebraic relation for $\mathcal{E}(t\mathbf{x})$ can be specialized to $t = 1$ after removing any common factor $t - 1$ from its coefficients. The specialization is valid coefficientwise in the $\mathbf{x}$ -adic topology and yields a nonzero polynomial relation for $\mathcal{E}(\mathbf{x})$. Alternatively (source 42), replace x_i by x_i/t in a polynomial relation for $\mathcal{E}(t\mathbf{x})$ and clear powers of t ; at least one coefficient in t is then a nonzero polynomial relation for $\mathcal{E}(\mathbf{x})$, and no relation that vanishes identically at $t = 1$ is ever specialized. $\square$

This is the classical directed-path kernel construction, included here to make the composition marking explicit; its general development is due to the directed-path literature, notably Banderier and Flajolet [8]. Source 42 gives the same proof for every finite list of labelled integer steps (Section 14).

4.3 Why D-finiteness follows

An algebraic multivariate formal series in characteristic zero is D-finite: its derivatives belong to its finite algebraic function field, so they span a finite-dimensional vector space over the rational function field. Lipshitz's diagonal theorem then implies that the complete diagonal (4.2) is D-finite [9].

Equivalently, for each fixed r there exist polynomials $q_0(n), \dots, q_d(n) \in \mathbb{Q}[n]$, not all zero, with

$$\sum_{j=0}^d q_j(n) A_r(n+j) = 0$$

for all sufficiently large n . This proves the D-finiteness portion of Theorem 3.2. Closure algorithms make this an effective existence result, but they do not imply a small order, a small degree, or a fast calculation of a useful recurrence. Source 42 proves the same statement by the same argument and notes why it does not conflict with the growing transfer matrices of [7]: a transfer matrix whose size increases with n does not have a fixed rational resolvent, but that observation is not an obstruction to a fixed *algebraic* multivariate series.

4.4 A concrete algebraic certificate for the fifth alphabet

For steps $-2, -1, 0, 1, 2$, give the weights the names a, b, c, d, e respectively. The ambient excursion series $X = \mathcal{E}(a, b, c, d, e)$ is the unique formal solution with constant coefficient one of

$$\begin{aligned} 0 = & a^3 e^3 X^6 + a^2 e^2 (c-1) X^5 + (abde - a^2 e^2) X^4 \\ & + (ad^2 + b^2 e + 2ae(1-c)) X^3 + (bd - ae) X^2 + (c-1) X + 1. \end{aligned} \quad (4.5)$$

Here the zero-step weight c is a variable, not the OEIS asymptotic constant.

To verify the equation, factor the quartic kernel as

$$e(u^2 - vu + q)(u^2 - wu + q') = eu^4 + du^3 + (c-1)u^2 + bu + a.$$

The two small roots have product $q = -aX$, while $q' = -1/(eX)$. Comparing the cubic and linear coefficients gives

$$v + w = -d/e, \quad vq' + wq = -b/e.$$

Solve these two equations for v, w , and substitute into $q + q' + vw = (c-1)/e$. Clearing denominators gives (4.5). This calculation is valid first for generic weights, and hence as a polynomial identity. At $a = b = c = d = e = 0$ the equation reduces to $1 - X = 0$, with nonzero derivative in X ; the formal implicit function theorem therefore selects a unique branch. The fifth row is the complete diagonal of this explicitly specified algebraic branch.

Remark 4.2 (Source 42's derivation). Source 42 obtains the same equation with the opposite overall sign, written in the abbreviations ae , bd and $ad^2 + b^2e$, by solving the coefficient equations explicitly. With the small-root factor $u^2 - vu + q$ and $q = -aX$ as above,

$$q' = -\frac{1}{eX}, \quad v = \frac{X(adX + b)}{1 - aeX^2}, \quad w = -\frac{d}{e} - v,$$

and substitution into the quadratic coefficient gives

$$[u^2] e(u^2 - vu + q)(u^2 - wu + q') - (c - 1) = -\frac{S(X)}{X(aeX^2 - 1)^2},$$

where $S(X)$ is the right side of (4.5); source 42's program checks this identity exactly. The sextic is a finite algebraic input for creative telescoping. It is not a certificate that a guessed diagonal operator annihilates the diagonal; that further elimination, followed by initial-condition checking, remains a concrete computational target (Section 22). (2026-10-02 note: the guessed operator for A215570 is now proved by another route, which needs no elimination from the sextic: Theorem 26.1. A third derivation of the sextic is Remark 25.1.)

5 The bridge logarithm and the algebraic constant

5.1 An exact identity retaining every composition

Define the bridge generating function

$$B(t; \mathbf{x}) = \text{CT}_u \frac{1}{1 - tP_{\mathbf{x}}(u)}.$$

Unlike excursions, bridges have no sign restriction on their intermediate heights. Their composition coefficients are multinomial coefficients.

Lemma 5.1 (Bridge logarithm). *As formal power series,*

$$B(t; \mathbf{x}) = 1 + t \frac{d}{dt} \log \mathcal{E}(t\mathbf{x}), \quad (5.1)$$

and consequently

$$\log \mathcal{E}(\mathbf{x}) = \sum_{\substack{\mathbf{k} \neq \mathbf{0} \\ \sum_i s_i k_i = 0}} \frac{(|\mathbf{k}| - 1)!}{\prod_i k_i!} \mathbf{x}^{\mathbf{k}}. \quad (5.2)$$

In particular,

$$[x_1^n \cdots x_r^n] \log \mathcal{E}(\mathbf{x}) = \frac{1}{rn} \frac{(rn)!}{(n!)^r} \quad (n \geq 1). \quad (5.3)$$

Proof. For small positive numerical weights and sufficiently small t , constant term extraction is a contour integral. Its residues at the small kernel roots give

$$B(t; \mathbf{x}) = \sum_{j=1}^h \frac{-1}{tu_j P'_{\mathbf{x}}(u_j)} = t \sum_{j=1}^h \frac{u'_j(t)}{u_j(t)}.$$

The second equality follows by differentiating $tP_{\mathbf{x}}(u_j(t)) = 1$. Taking the logarithmic derivative of (4.4) proves (5.1). The identity extends formally to arbitrary weights.

Expanding B in powers of t and then expanding $P_{\mathbf{x}}(u)^m$ gives the coefficient $m!/\prod k_i!$ precisely when $|\mathbf{k}| = m$ and $\sum s_i k_i = 0$. Integrating $(B - 1)/t$ supplies the factor $1/m$ and proves (5.2). Its balanced specialization gives (5.3). $\square$

The bridge–excursion relation is classical [8]. Its usefulness here is that the diagonal of the logarithm is *known exactly*. We can calibrate a multivariate saddle calculation against that coefficient and thereby avoid an unjustified guess about cyclic rotations or lattice periods.

Second proof of Lemma 5.1 (source 42). Expanding the logarithm gives

$$\mathrm{CT}_u[-\log(1 - tP_{\mathbf{x}}(u))] = \sum_{m \geq 1} \frac{t^m}{m} [u^0] P_{\mathbf{x}}(u)^m,$$

and $[u^0]P_{\mathbf{x}}(u)^m$ is precisely the multinomial generating polynomial for zero-sum words of length m . So this constant term is the right side of (5.2) at $t\mathbf{x}$, and it remains to identify it with $\log \mathcal{E}(t\mathbf{x})$; (5.1) then follows by differentiation. Factor the kernel in an annulus between its small and large roots:

$$1 - tP_{\mathbf{x}}(u) = \Theta(t) \prod_{j=1}^h (1 - u_j/u) \prod_{j=1}^h (1 - u/\tilde{u}_j),$$

where $\tilde{u}_1, \dots, \tilde{u}_h$ are the large roots. The coefficient of u^{-h} gives $\Theta(t)(-1)^h \prod_j u_j = -tx_1$; by Lemma 4.1, $\Theta = 1/\mathcal{E}(t\mathbf{x})$. The logarithms of the small-root factors have only negative powers of u , and those of the large-root factors have only positive powers. Taking the Laurent constant term therefore gives $\mathrm{CT}_u[-\log(1 - tP_{\mathbf{x}})] = -\log \Theta = \log \mathcal{E}(t\mathbf{x})$. One may first work at generic weights and then use formal coefficient identities to specialize them. All the series have the prescribed value at $t = 0$, fixing the logarithmic branch. $\square$

The identity is also the composition-marked case of a corollary proved in the collection's report on multiple-chain preorder polytopes over any commutative coefficient ring (Section 1.4).

5.2 The critical value

Set $\mathbf{p} = (1/r, \dots, 1/r)$. In the probability normalization,

$$B(t; \mathbf{p}) = \sum_{m \geq 0} \mathbb{P}(S_m = 0) t^m.$$

A finite-support, nondegenerate centered lattice walk has $\mathbb{P}(S_m = 0) = O(m^{-1/2})$. For completeness, this bound follows by Fourier inversion: the characteristic function has quadratic modulus decay near the finitely many points where its modulus is one, and has modulus bounded strictly below one on their complement. The integral near each such point is bounded by a Gaussian integral. Thus (3.2) converges. By monotone convergence in the nonnegative coefficient series,

$$\kappa_r = \lim_{t \uparrow 1} \mathcal{E}(t\mathbf{p}) = \mathcal{E}(\mathbf{p}) > 1. \quad (5.4)$$

Proposition 5.2 (Finite algebraic formula). *Let*

$$Q_r(u) = \sum_{i=1}^r u^{s_i+h} - ru^h. \quad (5.5)$$

The root $u = 1$ has multiplicity exactly two. Apart from that root, Q_r has $h - 1$ roots strictly inside the unit disk and $h - 1$ strictly outside, counting multiplicities. If the inside roots are $\alpha_1, \dots, \alpha_{h-1}$, then

$$\kappa_r = (-1)^{h+1} r \prod_{j=1}^{h-1} \alpha_j. \quad (5.6)$$

An empty product is one.

Proof. At $u = 1$, $P_r(1) = 1$, $P'_r(1) = 0$, and $P''_r(1) = \sum_i p_i s_i^2 > 0$. Hence the multiplicity is exactly two. If $|u| = 1$ and $P_r(u) = 1$, equality in the triangle inequality forces $u^{s_i} = 1$ for every i . The normalized step set has gcd one, so $u = 1$. There are no other roots on the unit circle.

For $0 < t < 1$, Rouché's theorem applied on $|u| = 1$ shows that $u^h(1 - tP_r(u))$ has exactly h roots inside. Near $t = 1$, its two roots near one split as $1 \pm c\sqrt{1-t} + O(1-t)$ with $c > 0$; one is inside and one outside. This leaves $h - 1$ strictly inside in the limit. Formula (5.6) is now the limit of (4.4). It also proves algebraicity. Positivity was already obtained from (5.4). $\square$

Internal multiple roots, should they occur, cause no difficulty: the product of a separated cluster is analytic even when individual root labels cannot be chosen analytically.

Source 42 proves the same formula as the limit at $t = 1$ of the small-root formula, with the finite value (3.2) identifying the branch and proving positivity. It counts the inside roots without Rouché's theorem: Q_r is a reciprocal polynomial of degree $2h$ with a double root at $u = 1$ and no other root on the unit circle, so reciprocity leaves $h - 1$ roots inside the unit disk. The prescription is exact; numerical root selection is only one way to evaluate it, and certified algebraic root isolation can replace floating-point root selection wherever a numerical interval certificate is required.

For the symmetric step sets in this paper one also has the real integral

$$\log \kappa_r = -\frac{1}{2\pi} \int_0^{2\pi} \log(1 - P_r(e^{i\theta})) d\theta. \quad (5.7)$$

Indeed, expand first with a factor $t < 1$, integrate term by term, and let $t \uparrow 1$. The logarithmic singularity at $\theta = 0$ is integrable. The integrand is real because $P_r(e^{i\theta})$ is real and at most one. Equation (5.7) is an alternative numerical expression, not a replacement for the algebraic proof.

6 Fixed-composition singularity analysis

Source 42 proves the expansion of this section by the same strategy in lattice coordinates, with an explicit Gaussian-moment algorithm (Section 17), and proves its leading term without any saddle point (Theorem 16.1).

6.1 Removing the two linear constraints

Let $d = r - 2$ and mark only the first d step types in addition to total length. Use logarithmic markers $\mathbf{y} = (y_1, \dots, y_d)$ and set

$$P_{\mathbf{y}}(u) = \sum_{i=1}^r p_i e^{y_i^*} u^{s_i}, \quad y_i^* = \begin{cases} y_i & i \leq d, \\ 0 & i > d. \end{cases} \quad (6.1)$$

If total length is N and the first d counts are known, the remaining two counts are uniquely determined by length and total height zero, because $s_{r-1} \neq s_r$. At $N = rn$, extracting n occurrences of each marked type therefore extracts the complete balanced composition.

The relevant coefficient functional is

$$\mathcal{C}_N(U) = [t^N z_1^n \cdots z_d^n] U(tp_1 z_1, \dots, tp_d z_d, tp_{r-1}, tp_r), \quad N = rn. \quad (6.2)$$

For $d = 0$ the z extraction is absent. Thus

$$\mathcal{C}_N(\mathcal{E}) = r^{-N} A_r(n), \quad \mathcal{C}_N(\log \mathcal{E}) = r^{-N} \frac{(rn)!}{rn(n!)^r}.$$

6.2 The moving critical point and its Hessian

The analytic implicit function theorem gives a unique $\tau(\mathbf{y})$ near one solving

$$(u\partial_u)P_{\mathbf{y}}(u)|_{u=\tau(\mathbf{y})} = 0, \quad \tau(0) = 1. \quad (6.3)$$

The required derivative in $\log u$ is the positive variance

$$V = \sum_i p_i s_i^2.$$

Define

$$\begin{aligned} \rho(\mathbf{y}) &= P_{\mathbf{y}}(\tau(\mathbf{y}))^{-1}, \\ \psi(\mathbf{y}) &= \log P_{\mathbf{y}}(\tau(\mathbf{y})), \\ \Phi(\mathbf{y}) &= \psi(\mathbf{y}) - \sum_{a=1}^d p_a y_a. \end{aligned} \quad (6.4)$$

Then $\nabla\Phi(0) = 0$. Its Hessian is

$$H_{ab} = p_a \delta_{ab} - p_a p_b - \frac{p_a s_a p_b s_b}{V} \quad (1 \leq a, b \leq d). \quad (6.5)$$

Lemma 6.1 (Nondegenerate composition saddle). *The matrix H is positive definite, and, with $\delta = s_r - s_{r-1} = 2/g_r$,*

$$\det H = \frac{\delta^2}{V} \prod_{i=1}^r p_i. \quad (6.6)$$

For $d = 0$, the determinant convention is $\det H = 1$.

Proof. For numbers $v_1, \dots, v_d$, extend $f_i = v_i$ for $i \leq d$ by $f_{r-1} = f_r = 0$. Direct differentiation in (6.3) gives

$$v^T H v = \text{Var}_p(f) - \frac{\text{Cov}_p(f, s)^2}{V}.$$

Cauchy–Schwarz makes this nonnegative. Equality would require $f_i = a + b s_i$ for every i . The two distinct steps with $f_i = 0$ force $a = b = 0$, hence $v = 0$.

For the determinant, let the random vector X have components $(\mathbf{1}_{\{i=1\}}, \dots, \mathbf{1}_{\{i=d\}}, s_i)$ with probability p_i . Its covariance matrix has Schur complement H in the last entry V , so its determinant is $V \det H$. A random vector supported on r affinely independent points in dimension $r - 1$ has covariance determinant equal to the product of its r probabilities times the square of the affine-simplex determinant. This identity follows by adjoining a row of ones to the support matrix and applying the Schur complement to its weighted Gram matrix. Here that affine determinant is, up to sign, $s_r - s_{r-1} = \delta$. This proves (6.6). $\square$

6.3 A uniform square-root expansion

Put

$$q_i(\mathbf{y}) = \frac{p_i e^{y_i^* \tau(\mathbf{y}) s_i}}{P_{\mathbf{y}}(\tau(\mathbf{y}))}, \quad V(\mathbf{y}) = \sum_i q_i(\mathbf{y}) s_i^2. \quad (6.7)$$

The q_i sum to one and have mean step zero. Multiplying weights by τ^{s_i} does not change any excursion weight, so the critical value is

$$\kappa(\mathbf{y}) = \mathcal{E}(q(\mathbf{y})).$$

It is analytic near $\mathbf{y} = 0$ as a critical root-cluster product, and $\kappa(0) = \kappa_r$.

Let $x = 1 - t/\rho(\mathbf{y})$. The small root which reaches the critical point has the expansion

$$u_0(t, \mathbf{y}) = \tau(\mathbf{y}) \left(1 - c(\mathbf{y})x^{1/2} + O(x) \right), \quad c(\mathbf{y}) = \sqrt{\frac{2}{V(\mathbf{y})}}.$$

All the other small roots are a separated cluster. Their product, and the factor $1/t$ in the kernel formula, are analytic at the critical point. Therefore, uniformly for complex $\mathbf{y}$ sufficiently near zero,

$$\mathcal{E}(t; \mathbf{y}) = \kappa(\mathbf{y}) \left(1 - c(\mathbf{y})x^{1/2} + O(x) \right), \quad (6.8)$$

$$\mathcal{L}(t; \mathbf{y}) := \log \mathcal{E}(t; \mathbf{y}) = \log \kappa(\mathbf{y}) - c(\mathbf{y})x^{1/2} + O(x). \quad (6.9)$$

Each expansion continues to arbitrary order in $x^{1/2}$. The notation $\mathcal{E}(t; \mathbf{y})$ means the marked specialization in (6.2), with $z_a = e^{y_a}$.

The square-root coefficients differ by exactly $\kappa(\mathbf{y})$. This analytic fact is the reason that the exact logarithmic reference coefficient in (5.3) identifies the fixed-composition leading constant.

6.4 Why other phases do not contribute

An asymptotic proof must control the full marker torus, not just a local expansion. Here this control is particularly simple.

Suppose $|t| = |u| = |z_a| = 1$ and a kernel root lies on $|u| = 1$. Equality in

$$\left| t \sum_i p_i z_i^* u^{s_i} \right| \leq 1$$

forces $tz_i^* u^{s_i} = 1$ for every i . The last two unmarked types imply $u^\delta = 1$. All step differences are multiples of δ , so then $z_a = 1$ for every marked type. If r is odd, $\delta = 1$ and $t = u = 1$. If r is even, $\delta = 2$ and the two possibilities are $(t, u) = (1, 1)$ and $(-1, -1)$.

Thus the only marker saddle is $\mathbf{z} = \mathbf{1}$. The only length period is

$$\nu = \begin{cases} 1 & r \text{ odd,} \\ 2 & r \text{ even,} \end{cases} \quad \text{so that } \nu = \delta. \quad (6.10)$$

In the even case every excursion has even length, and the two length singularities give identical contributions on the admissible subsequence $N = rn$.

More explicitly, on any compact part of the marker torus separated from $\mathbf{1}$, no kernel root can cross $|u| = 1$ for $|t| \leq 1$. By compactness this remains true for $|t| < 1 + \epsilon$ with some uniform $\epsilon > 0$. The small-root product is analytic there, even if roots within that cluster collide. It is nonzero away from $t = 0$, since the kernel has nonzero constant term for $t \neq 0$. The removable value at $t = 0$ is one. Consequently both $\mathcal{E}$ and its logarithm have uniform analytic continuations on that disk. Cauchy estimates make this part of the coefficient integral exponentially smaller than the main term.

6.5 Transfer and the Gaussian integral

For a half-integer γ which is not a nonnegative integer,

$$[t^N](1 - t/\rho)^\gamma = \rho^{-N} \frac{\Gamma(N - \gamma)}{\Gamma(-\gamma)\Gamma(N + 1)}. \quad (6.11)$$

The gamma quotient has a complete inverse-power expansion. Applying (6.11) to (6.8)–(6.9) gives local leading amplitudes

$$g_L(\mathbf{y}) = \frac{c(\mathbf{y})}{2\sqrt{\pi}}, \quad g_E(\mathbf{y}) = \kappa(\mathbf{y})g_L(\mathbf{y}), \quad (6.12)$$

multiplying $\rho(\mathbf{y})^{-N}N^{-3/2}$ and the period factor ν .

For marker extraction put $\mathbf{y} = i\boldsymbol{\theta}$. The exponential in the Cauchy integral is

$$\exp(N\Phi(i\boldsymbol{\theta})), \quad \Phi(i\boldsymbol{\theta}) = -\frac{1}{2}\boldsymbol{\theta}^\top H \boldsymbol{\theta} + O(|\boldsymbol{\theta}|^3).$$

The Hessian is positive definite. The leading Gaussian integral thus gives

$$\mathcal{C}_N(\mathcal{L}) \sim \frac{\nu g_L(0)}{(2\pi)^{d/2}\sqrt{\det H}} N^{-(d+3)/2}, \quad (6.13)$$

$$\mathcal{C}_N(\mathcal{E}) \sim \kappa_r \frac{\nu g_L(0)}{(2\pi)^{d/2}\sqrt{\det H}} N^{-(d+3)/2}. \quad (6.14)$$

The ratio is therefore κ_r . Substituting the exact logarithmic coefficient proves (3.5).

There is also a useful independent normalization check. From (6.6) and (6.10),

$$\frac{\nu g_L(0)}{(2\pi)^{d/2}\sqrt{\det H}} = \frac{\nu}{\delta} \frac{1}{(2\pi)^{(r-1)/2}\sqrt{\prod_i p_i}} = \frac{r^{r/2}}{(2\pi)^{(r-1)/2}}.$$

This is exactly the leading Stirling coefficient of $r^{-N}(rn)!/(rn(n!)^r)$ when $N = rn$. In particular, no hidden period-two factor is lost in the even alphabets.

6.6 Remainders and all orders

We spell out why the preceding calculation proves more than a leading heuristic. Fix any requested order M . The local kernel factorization provides Puiseux expansions to an arbitrarily high finite order, with analytic coefficient functions and uniform remainder estimates on a common dented neighborhood of the moving singularity. This follows from analytic factorization of the double root and analytic separation of the other root clusters. The same neighborhoods work for sufficiently small complex marker values.

After the length coefficient is extracted, Taylor-expand its analytic amplitudes and the phase Φ to a finite order larger than that needed for N^{-M-1} . In a neighborhood $|\boldsymbol{\theta}| \leq N^{-1/2+\varepsilon}$, substitute $\boldsymbol{\theta} = \mathbf{x}/\sqrt{N}$. Taylor's theorem and the positive-definite Gaussian majorant bound every discarded term by an integrable polynomial times a Gaussian, with the specified inverse power of N . One may first fix a sufficiently small $\varepsilon > 0$ depending on the requested truncation and then take sufficiently many Taylor terms. In the remainder of the small marker neighborhood, the real part of the phase is at most $-c|\boldsymbol{\theta}|^2$, giving an error smaller than every power of N^{-1} . The compact complement was handled by the analytic-continuation argument above.

Gaussian moments of odd total degree vanish. The half-integer powers in the length expansion differ by integers after coefficient transfer. Consequently only integer powers of N^{-1} remain after the common factor $N^{-(r+1)/2}$ is removed. This proves a full expansion with an $O(N^{-M-1})$ relative remainder after M corrections.

All finite Taylor jets involved are algebraic. Indeed, after replacing e^{y_a} by independent variables, the critical point solves polynomial kernel and derivative equations with algebraic base values. Exponential reparametrization has rational Taylor coefficients. Half-integer gamma

factors contribute one common $\sqrt{\pi}$, and Gaussian moments contribute rational expressions in H^{-1} times a common determinant factor. After division by the leading coefficient, every correction is algebraic.

Finally, Stirling’s expansion in (3.5), or directly (6.14), yields (3.3) and (3.4). This proves Theorem 3.1, including its all-order and algebraicity assertions.

Remark 6.2 (What “all orders” means here). Equation (3.3) determines the dominant exponential–power sector with a proved finite-order remainder for every fixed truncation. It does not assert convergence of the inverse-power series, identify every exponentially small contribution, or establish Borel summability. Those are separate questions. Source 42 adds that optimal truncation and a Gevrey bound are likewise not proved: the expansion is the dominant exponential sector of an asymptotic transseries, and its exponentially smaller sectors require additional global singularity analysis.

7 Explicit leading constants

The following values are obtained from the finite product, not from fitting a growth curve; Table 3 merges the tables of the two sources.

r	κ_r	$c(r)$ in (2.2)	$2\kappa_r - 1$
2	2.000000000000000	1.000000000000000	3.000000000000000
3	3.000000000000000	0.866025403784439	5.000000000000000
4	1.384057356943304	0.244669085661435	1.768114713886607
5	1.909830056250526	0.213525491562421	2.819660112501052
6	1.239095083832872	0.089423985025306	1.478190167665743
7	1.583896616939167	0.074832081265314	2.167793233878334
8	1.173744890362129	0.036679527823817	1.347489780724258
9	1.429003292368548	0.029770901924345	1.858006584737097
10	1.136489993032388	0.015882930501054	1.272979986064777
11	1.338806843936432	0.012614545363912	1.677613687872865
12	1.112404234469499	0.007095892609796	1.224808468938998

Table 3: Leading constants. The rows $r \leq 8$ of the first two columns were computed in both sources and agree to every digit either prints (source 61 printed 15 decimals, source 42 twelve); the rows $9 \leq r \leq 12$ and the last column, the limiting mean number of arches of Theorems 10.1 and 19.2, are source 42’s (its printed table stops at $r = 10$; its data file `data/42-late-growing-constants.csv` reaches $r = 12$, and those values are rounded here to 15 decimals). Decimal values are diagnostics, not interval-certified enclosures.

The printed decimals are numerical evaluations; the exact definition is (5.6). The evident even–odd pattern suggests a separate large-alphabet problem (source 42). None of the fixed- r proofs establishes a uniform two-parameter expansion.

For $r = 2, 3$, $h = 1$, so the product is empty and $\kappa_2 = 2$, $\kappa_3 = 3$. Directly,

$$A_2(n) = \frac{(2n)!}{n!(n+1)!}, \quad A_3(n) = \frac{(3n)!}{(n!)^3(n+1)}.$$

For the second formula, remove the n zero steps from a balanced Motzkin excursion, obtaining a Dyck path with n up and n down steps; reinsert the zero steps in $\binom{3n}{n}$ ways; source 42 derives both formulas the same way. These formulas give the checks

$$\alpha_{2,1} = -\frac{9}{8}, \quad \alpha_{3,1} = -\frac{11}{9}.$$

For the fourth row,

$$Q_4(u) = (u-1)^2(u^4 + 2u^3 + 4u^2 + 2u + 1).$$

The quartic is reciprocal. On dividing by u^2 and putting $v = u + u^{-1}$, it becomes $v^2 + 2v + 2 = 0$. Pair the two roots strictly inside the unit disk and multiply them. If $\varphi = (1 + \sqrt{5})/2$, the result is

$$\kappa_4 = 4(\varphi - \sqrt{\varphi}), \quad c(4) = \frac{\varphi - \sqrt{\varphi}}{\sqrt{2}}.$$

Substitution in (3.3) reproduces the formula credited in the OEIS entry to Kotesovec [2]. Source 42 reaches the same value by elimination: write the quartic as $(u^2 - \sigma u + q)(u^2 - (\sigma/q)u + q^{-1})$, where q is the product of its two interior roots. Comparing cubic and quadratic coefficients gives $\sigma = -2q/(1+q)$ and $q + q^{-1} + \sigma^2/q = 4$; eliminating σ gives $q^4 - 2q^3 - 2q^2 - 2q + 1 = 0$, and the root selected by $0 < q < 1$ has $q + q^{-1} = 1 + \sqrt{5} = 2\varphi$, so that $\kappa_4 = 4q$ is the value above.

For the fifth row,

$$Q_5(u) = (u-1)^2(u^2 + 3u + 1).$$

The unique strict inside root is $\alpha = (-3 + \sqrt{5})/2$. Hence

$$\kappa_5 = \frac{5}{2}(3 - \sqrt{5}), \quad c(5) = \frac{3\sqrt{5} - 5}{8}.$$

This is the leading constant already recorded in A215570 [3]. As source 42 puts it, the claim for $r = 4, 5$ is a proof and a general prescription, not that these two radical expressions were previously absent. For reproducibility, two larger reduced kernels are

$$\begin{aligned} \frac{Q_6(u)}{(u-1)^2} &= u^8 + 2u^7 + 4u^6 + 6u^5 + 9u^4 + 6u^3 + 4u^2 + 2u + 1, \\ \frac{Q_7(u)}{(u-1)^2} &= u^4 + 3u^3 + 6u^2 + 3u + 1. \end{aligned}$$

Together with the inside-root selection, these give exact algebraic specifications of the constants for A215571 and A215593, whose checked entries did not give asymptotic formulas [4, 5].

8 Why every nontrivial higher row is transcendental

The exponent and constant in Theorem 3.1 imply a complete algebraicity classification for the row generating functions.

Proof of the transcendence assertion in Theorem 3.2. Set $R = r^{-r}$ and $G(x) = F_r(Rx)$. Its coefficients satisfy

$$[x^n]G(x) \sim K_r n^{-(r+1)/2}, \quad K_r > 0.$$

Assume for contradiction that G is algebraic over $\mathbb{Q}(x)$. Every ordinary derivative of G is then algebraic and, near the algebraic point $x = 1$, has a Puiseux expansion with algebraic coefficients.

First let $r = 2k + 1 \geq 3$. Then

$$G^{(k)}(x) \sim K_r \log \frac{1}{1-x} \quad (x \uparrow 1).$$

To justify this Abelian estimate, its coefficient sequence is asymptotic to K_r/n up to the harmless derivative index shift. Compare it, outside any fixed finite initial segment, with

$(K_r \pm \epsilon)/n$ and use $\sum_{n \geq 1} x^n/n = -\log(1-x)$. No Puiseux series can have a leading logarithmic divergence. This is a contradiction.

Next let $r = 2k \geq 4$. Differentiation k times gives coefficients asymptotic to $K_r n^{-1/2}$. The same comparison argument and the Gaussian integral, or a Riemann-sum estimate with $x = e^{-t}$, yield

$$G^{(k)}(x) \sim K_r \sqrt{\pi} (1-x)^{-1/2}. \quad (8.1)$$

The leading Puiseux coefficient of an algebraic function over $\mathbb{Q}(x)$ at $x = 1$ must be algebraic. But

$$K_r \sqrt{\pi} = \frac{\kappa_r}{\sqrt{r} 2^{k-1/2}} \pi^{-(k-1)}$$

is transcendental: its prefactor is nonzero algebraic, and $k-1$ is a positive integer. If this number were algebraic, then π^{k-1} and hence π would be algebraic, contradicting Lindemann's theorem [10]. This contradicts (8.1).

Finally $F_2(z) = (1 - \sqrt{1-4z})/(2z)$ is algebraic, as expected. $\square$

The argument is not a generic claim that all diagonals are transcendental. It uses the particular positive leading asymptotic and its algebraic constant. It also explains why the even rows require an arithmetic argument in addition to their half-integer growth exponent.

9 The first correction for the fifth row

9.1 A differential formula for the ratio correction

This section makes the all-order construction concrete. Its calculation is independent of the conjectured A215570 recurrence (proved in Part IV, which uses the value of $\alpha_{5,1}$ found here as an initial datum).

Use the phase and marker notation from Section 6, and let $C = H^{-1}$. Repeated indices in the formulas below are summed from 1 to d . Put

$$\ell(\mathbf{y}) = \log \kappa(\mathbf{y}), \quad h_c(\mathbf{y}) = \log c(\mathbf{y}), \quad T_{abc} = \partial_a \partial_b \partial_c \Phi(0).$$

All derivatives without an argument are evaluated at zero. For an analytic amplitude g , the order- N^{-1} contribution in its Gaussian saddle integral is

$$-\frac{1}{2} C_{ab} g_{ab} + \frac{1}{2} g_a T_{bcd} C_{ab} C_{cd} + g \mathcal{J},$$

where $\mathcal{J}$ is independent of the amplitude. This follows by expanding $g(i\mathbf{x}/\sqrt{N})$ and the cubic and quartic terms of $N\Phi(i\mathbf{x}/\sqrt{N})$, then using Gaussian moments. The signs come from $i^2 = -1$; in particular the second derivative term is negative.

Subtracting the normalized correction for g_L from that for $g_E = \kappa g_L$ cancels $\mathcal{J}$ and gives

$$\mathcal{S} = -\frac{1}{2} C_{ab} (\ell_{ab} + \ell_a \ell_b) - C_{ab} \ell_a (h_c)_b + \frac{1}{2} \ell_a T_{bcd} C_{ab} C_{cd}. \quad (9.1)$$

There is a second correction from the length singularity itself. Write at $\mathbf{y} = 0$

$$\begin{aligned} \mathcal{E}(t) &= \kappa_r (1 - cx^{1/2} + Dx + E_3 x^{3/2} + O(x^2)), \\ \log \mathcal{E}(t) &= \log \kappa_r - cx^{1/2} + L_2 x + L_3 x^{3/2} + O(x^2), \quad x = 1 - t. \end{aligned}$$

Then $D = L_2 + c^2/2$. In the coefficient transfer formula the universal $3/(8N)$ term cancels between the two functions, and $\Gamma(-1/2)/\Gamma(-3/2) = -3/2$. It follows that the difference of their relative length corrections is

$$\mathcal{B} = -\frac{3}{2}L_2 - \frac{c^2}{4}. \quad (9.2)$$

Combining the two extractions,

$$A_r(n) = \frac{\kappa_r (rn)!}{rn (n!)^r} \left(1 + \frac{\Delta_r}{rn} + O(n^{-2})\right), \quad \Delta_r = \mathcal{B} + \mathcal{S}. \quad (9.3)$$

Stirling's first correction then gives

$$\alpha_{r,1} = \frac{1/r - r}{12} + \frac{\Delta_r}{r}. \quad (9.4)$$

Equations (9.1)–(9.4) are also an implementable recipe for other fixed rows.

9.2 Exact ingredients for $r = 5$

Here $s = (-2, -1, 0, 1, 2)$, $p_i = 1/5$, $d = 3$, and $V = 2$. The Hessian and inverse are

$$H = \begin{pmatrix} 2/25 & -2/25 & -1/25 \\ -2/25 & 7/50 & -1/25 \\ -1/25 & -1/25 & 4/25 \end{pmatrix}, \quad C = \begin{pmatrix} 130 & 90 & 55 \\ 90 & 70 & 40 \\ 55 & 40 & 30 \end{pmatrix}.$$

The large entries in C are a consequence of marking the first three letters while reconstructing the final two. They are not evidence of a degenerate saddle: H is strictly positive definite.

For compact formulas define the centered residual markers

$$R_a(i) = \mathbf{1}_{\{i=a\}} - p_a - \frac{p_a s_a}{V} s_i. \quad (9.5)$$

Then

$$T_{abc} = \sum_i p_i R_a(i) R_b(i) R_c(i), \quad (h_c)_a = -\frac{1}{2V} \sum_i p_i s_i^2 R_a(i). \quad (9.6)$$

The cubic identity is the third derivative of the profiled log moment generating function in (6.4); derivatives of the optimized height coordinate cancel because each residual marker is orthogonal to the height step.

There is an especially simple formula for $\kappa(\mathbf{q})$ for nearby centered five-step probabilities. Write

$$A = q_{-2}, \quad E = q_2, \quad B = q_1 + 2q_2 = q_{-1} + 2q_{-2}.$$

The equality for B is the zero-drift condition. The critical kernel factors as

$$\sum_{s=-2}^2 q_s u^{s+2} - u^2 = (u-1)^2 (Eu^2 + Bu + A),$$

so

$$\kappa(\mathbf{q}) = \frac{2}{B + \sqrt{B^2 - 4AE}}. \quad (9.7)$$

The square root is positive at the uniform probabilities and continued analytically nearby.

For transparency, let $v = \log \tau$. Differentiating the zero-drift equation and (6.7) gives

$$\begin{aligned} v_a &= -p_a s_a / V, & v_{ab} &= -V^{-1} \sum_i p_i s_i R_a(i) R_b(i), \\ (q_i)_a &= p_i R_a(i), & (q_i)_{ab} &= p_i (R_a(i) R_b(i) + s_i v_{ab} - H_{ab}). \end{aligned} \quad (9.8)$$

Equations (9.7)–(9.8) determine the needed derivatives using only rational arithmetic and $\sqrt{5}$. Their values are

$$(\ell_a) = \left(-\frac{3}{10} + \frac{\sqrt{5}}{10}, \quad \frac{1}{5} - \frac{\sqrt{5}}{10}, \quad \frac{1}{5} \right), \quad ((h_c)_a) = \left(-\frac{1}{10}, \frac{1}{20}, \frac{1}{10} \right),$$

and

$$(\ell_{ab}) = \begin{pmatrix} \frac{2}{25} - \frac{4\sqrt{5}}{125} & -\frac{2}{25} + \frac{11\sqrt{5}}{250} & -\frac{1}{25} \\ -\frac{2}{25} + \frac{11\sqrt{5}}{250} & \frac{7}{50} - \frac{17\sqrt{5}}{250} & -\frac{1}{25} \\ -\frac{1}{25} & -\frac{1}{25} & \frac{4}{25} \end{pmatrix}.$$

Substitution in (9.1), using (9.6), now gives the exact contraction

$$\mathcal{S} = -\frac{11}{4} + \sqrt{5}. \quad (9.9)$$

To obtain L_2 , note that the uniform law is symmetric. The logarithm of the critical small root has no x term: in the coordinate $u = e^w$, $P(e^w)$ is even in w , so $w = -c\sqrt{x} + O(x^{3/2})$. The other small root is $\alpha = (-3 + \sqrt{5})/2$. Differentiating its kernel equation yields

$$L_2 = 1 + \frac{1}{\alpha P'_5(\alpha)} = 1 - \frac{\sqrt{5}}{5}.$$

Since $c^2 = 2/V = 1$, equation (9.2) gives

$$\mathcal{B} = -\frac{7}{4} + \frac{3\sqrt{5}}{10}, \quad \Delta_5 = -\frac{9}{2} + \frac{13\sqrt{5}}{10}.$$

Finally (9.4) gives

$$\alpha_{5,1} = -\frac{13}{10} + \frac{13\sqrt{5}}{50} = -\frac{13(5 - \sqrt{5})}{50}.$$

This proves (3.6).

All higher finite jets at the uniform point stay in $\mathbb{Q}(\sqrt{5})$: the critical root-cluster data have that field, $c = 1$, and all implicit recursions have nonzero denominators in the same field. Gaussian and Stirling corrections introduce only rational operations. This proves the field assertion in Theorem 3.3.

Source 42 obtains $\alpha_{5,1}$ independently from a different, coordinate-invariant contraction formula (Section 18), and recovers $\alpha_{3,1} = -11/9$ from that general formula rather than from the closed form. In the multinomial normalization (9.3), $\Delta_5/5 = -9/10 + 13\sqrt{5}/50$, which is source 42's (18.7).

10 A fixed-composition contact law

The same saddle comparison provides distributional information, not only total enumeration. An *arch* is a nonempty excursion with no interior return to height zero. A zero step at height

zero is itself an arch. Every nonempty excursion decomposes uniquely into a sequence of arches. Hence the arch generating function is

$$\mathcal{I}(\mathbf{x}) = 1 - \mathcal{E}(\mathbf{x})^{-1}.$$

Let J_n be the number of arches of a uniformly chosen balanced r -letter excursion, and let $C_n = J_n - 1$ be its number of interior returns to zero.

Theorem 10.1 (Contact limit under exact composition). *For each fixed $r \geq 2$ and $k \geq 0$,*

$$\mathbb{P}(C_n = k) \longrightarrow \frac{k+1}{\kappa_r^2} \left(1 - \frac{1}{\kappa_r}\right)^k. \quad (10.1)$$

The probability generating functions converge in a complex neighborhood of $v = 1$, and

$$\mathbb{E}C_n \longrightarrow 2(\kappa_r - 1). \quad (10.2)$$

Proof. For any analytic function f near κ_r , the leading square-root coefficient of $f(\mathcal{E})$ is $f'(\kappa_r)$ times that of $\mathcal{E}$. The same coefficient-extraction proof therefore gives the leading ratio $f'(\kappa_r)$, provided the composition is analytic on the necessary neighborhoods. For fixed k , apply this with $f(E) = (1 - E^{-1})^{k+1}$. It counts exactly $k+1$ arches, and the ratio is

$$f'(\kappa_r) = \frac{k+1}{\kappa_r^2} (1 - \kappa_r^{-1})^k.$$

These nonnegative limiting probabilities sum to one.

For the stronger generating-function statement, mark arches by v :

$$\mathcal{E}_v = \frac{1}{1 - v\mathcal{I}}.$$

At the critical weights, $\mathcal{I} = \eta := 1 - \kappa_r^{-1} < 1$. On the closed coefficient polydisk, nonnegative coefficients give $|\mathcal{I}| \leq \eta$. For $|v| < 1/\eta$ the denominator is consequently bounded away from zero; this remains true in a suitable local continuation. The preceding analysis is therefore uniform for v near one. Its derivative ratio is

$$\mathbb{E}(v^{J_n}) \longrightarrow \frac{v}{\kappa_r^2(1 - \eta v)^2}.$$

Divide by v and differentiate at one to obtain (10.2). □

A related negative-binomial contact law is classical when only length is conditioned [8]. The statement here imposes the complete step composition. Its proof explains why the same critical value survives this additional conditioning.

Source 42 proves the same limit law, stated for the number $J_n = C_n + 1$ of arches (its primitive factors) and for every centred rational composition, together with convergence of all fixed moments, by a cutoff argument that uses no singularity analysis (Theorem 19.2); its limiting mean is $\mathbb{E}J_n \rightarrow 2\kappa_r - 1$.

11 Inverting the asymptotic expansion

11.1 An exact Lambert core

For fixed r define

$$\lambda = r \log r, \quad \beta = \frac{r+1}{2}, \quad K = K_r.$$

Then the proved expansion has the form

$$A_r(n) \sim K e^{\lambda n} n^{-\beta} \left(1 + \frac{\alpha_1}{n} + \frac{\alpha_2}{n^2} + \cdots \right). \quad (11.1)$$

The large real solution of the *leading model* $Y = K e^{\lambda x} x^{-\beta}$ is exactly

$$x_0(Y) = -\frac{\beta}{\lambda} W_{-1} \left(-\frac{\lambda}{\beta} \left(\frac{K}{Y} \right)^{1/\beta} \right). \quad (11.2)$$

The lower real branch W_{-1} is essential: it selects $x_0 \rightarrow \infty$. The argument lies in $(-e^{-1}, 0)$ for all sufficiently large Y . This exact-core organization parallels the inversion framework inspected in ProveIt [11]; no unverified asymptotic interpolation is needed. The exact core and its perturbation are the instance $(K, \lambda, \beta) = (K_r, r \log r, (r+1)/2)$ of the canonical volume's inversion apparatus (Section 1.4), and source 42 derives the same core and the same first corrections (Section 20); no novelty is claimed for them.

11.2 Corrections and a remainder statement

Write the formal logarithm of the unit correction as

$$\log \left(1 + \sum_{j \geq 1} \alpha_j x^{-j} \right) = \sum_{j \geq 1} b_j x^{-j}, \quad b_1 = \alpha_1, \quad b_2 = \alpha_2 - \frac{1}{2} \alpha_1^2.$$

Seek $x = x_0 + \sum_{j \geq 1} \mu_j x_0^{-j}$ in

$$\lambda(x - x_0) - \beta \log(x/x_0) + \sum_{j \geq 1} b_j x^{-j} = 0. \quad (11.3)$$

At every order the new coefficient occurs with nonzero multiplier λ , so the recursion is triangular. Its first terms are

$$x = x_0 - \frac{b_1}{\lambda x_0} - \frac{b_2/\lambda + \beta b_1/\lambda^2}{x_0^2} + O(x_0^{-3}). \quad (11.4)$$

This has a precise interpretation on the original sequence. Given any fixed order, evaluate the truncated inverse at $Y = A_r(n)$. The forward expansion has a controlled logarithmic residual, while the derivative of $\lambda x - \beta \log x$ is $\lambda - \beta/x$, bounded below by $\lambda/2$ for large x . The mean value theorem transports the residual to an error of the same inverse-power order in the recovered n . Thus the inversion is a theorem about the sequence values and their asymptotic inverse, rather than a claim that a canonical analytic interpolation of the integers has been constructed.

For the fifth row, (11.4) begins

$$n = x_0 + \frac{13(5 - \sqrt{5})}{250 \log 5} \frac{1}{x_0} + O(x_0^{-2}), \quad Y = A_5(n), \quad (11.5)$$

where $\beta = 3$, $\lambda = 5 \log 5$, and $K = (3\sqrt{5} - 5)/(8\pi^2)$ in (11.2). (2026-10-02 note: with $\alpha_{5,2}$ now known, the x_0^{-2} term is explicit, (28.1).)

11.3 Integer thresholds are a separate operation

For arbitrary real Y , define

$$N_r(Y) = \min\{n \geq 0 : A_r(n) \geq Y\}.$$

Appending a fixed balanced excursion block containing one of each step is an injection from the n th class into the $(n+1)$ st. Hence $A_r(n)$ is nondecreasing; its proved asymptotic also gives $A_r(n+1)/A_r(n) \rightarrow r^x > 1$, so it is eventually strictly increasing.

The continuous inverse predicts $N_r(Y)$ to bounded additive error. Taking a ceiling gives the exact threshold when an explicit inverse-error bound separates the predicted value from the relevant integers. Without that separation there is no universally valid exact rounding rule: Y can be arbitrarily close to a jump value $A_r(n)$. This is the discrete staircase issue emphasized in the repository's inversion documentation, and it must not be hidden in an $o(1)$ statement about a continuous model. In the canonical transseries volume it is the staircase theorem `p0:thm:staircase`, whose separation clause is formalized in Lean (Remark 20.1); source 42 gives an explicit separation bound, (20.4).

12 Unequal but centered compositions

The main construction is not restricted to equal multiplicities. Keep the step set (3.1), and let $p_1, \dots, p_r$ be fixed positive rational numbers with

$$\sum_i p_i = 1, \quad \sum_i p_i s_i = 0.$$

For admissible N , let $e(N\mathbf{p})$ count excursions with exactly Np_i occurrences of step i . Define $\kappa(\mathbf{p}) = \mathcal{E}(\mathbf{p})$ by the same critical kernel product, now using these probabilities.

Corollary 12.1 (Centered rational rays). *Along integers N for which all Np_i are integral,*

$$e(N\mathbf{p}) \sim \frac{\kappa(\mathbf{p})}{N} \frac{N!}{\prod_i (Np_i)!}. \quad (12.1)$$

There is a full inverse-power expansion in N with algebraic normalized coefficients. The coefficient series on every fixed positive integral ray is D -finite.

Proof. Replace the uniform probabilities everywhere in Section 6 by the specified p_i . Centering keeps the critical height coordinate at one. The Hessian proof, its determinant, and the marker-torus equality argument are unchanged. The exact coefficient of $\log \mathcal{E}$ is again the multinomial divided by N . Its ratio with the excursion coefficient gives (12.1), and the finite-jet argument gives all orders.

For the final statement, a fixed rational-ray extraction can be implemented by coefficient sections and a diagonal. Sections are finite root-of-unity filters after a monomial substitution; algebraicity is preserved by these operations, and the subsequent diagonal is D -finite. Equivalently one may use the multivariate D -finite residue closure for the fixed linear coefficient constraints. The dimensions and integer ray vector remain fixed. $\square$

This extension concerns excursions relative to the *same* height functional s_i . Centering ensures the final height is zero. Compositions with nonzero mean would describe a different endpoint problem and are not covered by this statement. Source 42 proves the leading term (12.1) for every finite list of labelled integer steps and every centred rational composition, by an elementary convolution argument (Theorem 16.1).

13 Reproducible checks and their scope

13.1 Exact finite enumeration

Source 61's program `code/61-balanced-verify.py` uses a direct composition recurrence. For a count vector $\mathbf{k}$, let $D(\mathbf{k})$ count nonnegative paths with that composition, regardless of final height. Then

$$D(\mathbf{0}) = 1, \quad D(\mathbf{k}) = 0 \text{ if } \sum_i s_i k_i < 0,$$

and for every other nonempty vector,

$$D(\mathbf{k}) = \sum_{i: k_i > 0} D(\mathbf{k} - \mathbf{e}_i).$$

The recurrence follows by deleting the last step. Its implementation stores a finite rectangular count-vector box in mixed-radix order, so every predecessor is already available. The diagonal gives $A_r(n)$.

The executed ranges were

r	Independently generated n	Rectangular states
2	0–24	625
3	0–24	15,625
4	0–18	130,321
5	0–16	1,419,857
6	0–6	117,649
7	0–4	78,125

The first two rows were compared with their factorial formulas. The other rows were compared with the available initial values in the corresponding OEIS entries. All comparisons passed. Source 42's independent dynamic program, with companion arrays for primitive words and returns, is described in Section 21; on the 47 indices that both programs computed ($r = 4, 5, 6, 7$ and $n \leq 18, 15, 6, 4$) the terms are equal.

An independent exact check of the bridge logarithm uses the identity $D_{\text{Euler}}\mathcal{E} = \mathcal{E}D_{\text{Euler}} \log \mathcal{E}$. Coefficientwise this is

$$|\mathbf{k}|e_{\mathbf{k}} = \sum_{\substack{0 < \mathbf{j} \leq \mathbf{k} \\ s: j=0}} \frac{|\mathbf{j}|!}{\prod_i j_i!} e_{\mathbf{k}-\mathbf{j}}. \quad (13.1)$$

Exact rational arithmetic in this identity agreed with the direct path recurrence for 120 nonempty balanced vectors across five test boxes. A separate height-based weighted enumeration verified the sextic (4.5) through length 24 at weights (2, 3, 5, 7, 11). The polynomial elimination proving that sextic is also reproduced in `code/61-balanced-derive_alpha5.py`.

13.2 Asymptotic and inverse diagnostics

For A_5 , write $L_5(n) = K_5 5^{5n} n^{-3}$ and $R_n = A_5(n)/L_5(n)$. The following values illustrate the independently derived first correction:

n	R_n	$n(R_n - 1)$	$n^2(R_n - 1 - \alpha_{5,1}/n)$
4	0.838539713679	-0.645841145284	0.291124722264
8	0.914884186065	-0.680926511482	0.301566514946
12	0.942234769960	-0.693182760477	0.305274784481
16	0.956285986890	-0.699424209762	0.307169857409
20	0.964839683966	-0.703206320687	0.308320103255
50	0.985752002111	-0.712399894449	0.311121570077

The predicted limit of the middle diagnostic is

$$\alpha_{5,1} = -0.7186223258500546789 \dots$$

The values for $n \leq 16$ were generated independently by the supplied program. The $n = 20, 50$ values were read from the published OEIS b-file; they were *not* independently enumerated here [6]. The last column is a numerical check of a second-order residual, not a claimed formula or a numerical certificate for $\alpha_{5,2}$. (2026-10-02 note: Part IV gives $\alpha_{5,2} = 36/25 - 63\sqrt{5}/125 = 0.31302\dots$ exactly, the limit of that column, Table 6; and its proved coefficient formula reproduces the $n = 20, 50$ rows, Theorem 25.2.)

At $Y = A_5(n)$, the Lambert core and its first correction give

n	$x_0(Y) - n$	$x_0(Y) - \alpha_{5,1}/(5 \log 5 x_0(Y)) - n$
8	$-1.1595249025 \cdot 10^{-2}$	$-4.1641746283 \cdot 10^{-4}$
16	$-5.6870504107 \cdot 10^{-3}$	$-1.0375143173 \cdot 10^{-4}$
50	$-1.7966866816 \cdot 10^{-3}$	$-1.0601877690 \cdot 10^{-5}$

The observed scales are consistent with the proved first- and second-order inverse errors (the last column is accounted for by the explicit x_0^{-2} term of (28.1)). Numerical agreement is a useful implementation check, not a substitute for the asymptotic proof.

13.3 What the computations do not certify

The package does not verify a Lean formalization, prove the guessed A215570 recurrence (source 75-03's program checks the certificate of Part IV, Section 29), find the minimal recurrence for any higher row, or supply interval-certified decimal enclosures for every kernel root. Its exact tests validate finite combinatorial and algebraic calculations. Its root computations and asymptotic tables are high-precision numerical diagnostics. The infinite statements are supported by the mathematical proofs above.

Part II

The convolution route, general step sets and the shape of an excursion

This Part prints the results of source 42 that source 61 does not contain, and source 42's proofs of shared results where they follow a genuinely different route. It uses the merged notation of Section 1.5; Table 2 lists every symbol of source 42 that was renamed. Statements proved in both sources are printed once, in Part I, and are only cited here.

14 Labelled steps and the log-excursion series

14.1 Labelled steps and composition

For a finite list of integer steps $\mathbf{s} = (s_1, \dots, s_q)$ having at least one negative and one positive entry, write

$$\mathcal{H} = \{\mathbf{k} \in \mathbb{N}^q : \mathbf{s} \cdot \mathbf{k} = 0\}, \quad |\mathbf{k}| = k_1 + \dots + k_q.$$

An excursion is a word in the *labelled* steps whose partial sums are nonnegative and whose final sum is zero. Distinct labels remain distinct even when their numerical steps coincide. Let $e_{\mathbf{k}}$ count excursions with composition $\mathbf{k}$, and set

$$\mathcal{E}(\mathbf{x}) = \sum_{\mathbf{k} \in \mathcal{H}} e_{\mathbf{k}} \mathbf{x}^{\mathbf{k}}, \quad e_{\mathbf{0}} = 1. \quad (14.1)$$

For the rows, $q = r$ and the steps are (3.1); the encoding of Section 4.1 turns the prefix condition into this nonnegative convention without changing the balanced composition, and (4.2) expresses $A_r(n)$ as a diagonal coefficient.

Lemma 4.1 holds for every such step list (source 42 states it in this generality): put $h = -\min_i s_i$, $h' = \max_i s_i$ and $x_- = \sum_{i:s_i=-h} x_i$, and replace x_1 by x_- in (4.4). The proof given there applies verbatim, because $u^h(1 - tP_{\mathbf{x}}(u))$ evaluated at $u = 0$ is $-tx_-$, and so do both arguments given there for removing the auxiliary variable. In particular $\mathcal{E}(\mathbf{x})$ is algebraic over $\mathbb{Q}(\mathbf{x})$. Generically the subset-product construction gives the finite degree bound $\binom{h+h'}{h}$, although the actual degree can be smaller.

14.2 The log-excursion series

Define

$$\mathcal{L}(\mathbf{x}) = \sum_{\substack{\mathbf{k} \in \mathcal{H} \\ |\mathbf{k}| > 0}} b_{\mathbf{k}}^0 \mathbf{x}^{\mathbf{k}}, \quad b_{\mathbf{k}}^0 = \frac{1}{|\mathbf{k}|} \binom{|\mathbf{k}|}{k_1, \dots, k_q}. \quad (14.2)$$

Lemma 5.1 and its proof, which uses only the kernel lemma, hold in this generality; in source 42's form the lemma reads

$$\mathcal{E}(\mathbf{x}) = \exp \mathcal{L}(\mathbf{x}), \quad \mathcal{L}(t\mathbf{x}) = \text{CT}_u [-\log(1 - tP_{\mathbf{x}}(u))]; \quad (14.3)$$

this is source 42's form of the bridge-excursion identity, and its own proof of it is printed after Lemma 5.1 as a second proof. The factor $1/|\mathbf{k}|$ is crucial. Replacing $\mathcal{L}$ by the unnormalized bridge series would give a false identity.

15 A directional convolution theorem

This section contains the elementary analytic mechanism behind the leading equivalent. It avoids assuming finite moments of the bridge-series mass; those moments generally diverge.

15.1 Critical multinomial weights

Fix positive rational numbers $p_1, \dots, p_q$ with

$$\sum_i p_i = 1, \quad \sum_i p_i s_i = 0, \quad \sum_i p_i s_i^2 > 0. \quad (15.1)$$

Let N tend to infinity through integers for which $\mathbf{k}_N = N\mathbf{p}$ is integral. Define

$$b_{\mathbf{k}} = b_{\mathbf{k}}^0 \mathbf{p}^{\mathbf{k}} \quad (\mathbf{k} \in \mathcal{H} \setminus \{0\}), \quad b_0 = 0, \quad \mathfrak{B} = \sum_{\mathbf{k}} b_{\mathbf{k}}.$$

Here and below sequences indexed off $\mathcal{H}$ are extended by zero. Put $\varsigma = (q+1)/2$.

Lemma 15.1 (Mass, bounds, and fixed shifts). *The mass $\mathfrak{B}$ is finite and positive. There are constants $c, C > 0$ such that*

$$b_{\mathbf{k}} \leq C|\mathbf{k}|^{-\varsigma} \quad (|\mathbf{k}| > 0), \quad b_{\mathbf{k}_N} \sim \frac{N^{-\varsigma}}{(2\pi)^{(q-1)/2} \sqrt{p_1 \cdots p_q}}. \quad (15.2)$$

For each fixed $\ell \in \mathcal{H}$,

$$\frac{b_{\mathbf{k}_N - \ell}}{b_{\mathbf{k}_N}} \longrightarrow 1. \quad (15.3)$$

The same conclusions along the target ray hold after any fixed admissible shift of that target.

Proof. Let Y take value s_i with probability p_i . Summing by length gives

$$\mathfrak{B} = \sum_{m \geq 1} \frac{\mathbb{P}(Y_1 + \cdots + Y_m = 0)}{m}.$$

The return probability is $O(m^{-1/2})$. One direct proof uses the Fourier integral of the characteristic function. Its modulus equals one at only finitely many points on a period, because the step distribution is nonconstant. Near each such point its modulus is at most $1 - c\theta^2$, after translating the point, and elsewhere it is bounded away from one. The Gaussian integral bounds the result by $Cm^{-1/2}$. The positive and negative steps and rational probabilities ensure a nonempty zero-sum word, so $\mathfrak{B} > 0$.

For the uniform pointwise bound in (15.2), the maximum point mass of a fixed q -category multinomial distribution is $O(m^{-(q-1)/2})$. To verify this directly, when every $k_i \geq p_i m/2$, Stirling's bounds give the stated power times $\exp(-mD_{\text{KL}}(\mathbf{k}/m \parallel \mathbf{p})) \leq 1$. Here $D_{\text{KL}}(\boldsymbol{\varpi} \parallel \mathbf{p}) = \sum_i \varpi_i \log(\varpi_i/p_i)$ is relative entropy, with $0 \log 0 = 0$. On the complementary closed subset of the probability simplex the relative entropy is bounded below by a positive constant. Factorial bounds, or the multinomial theorem followed by that entropy bound, give an exponentially small term, which is smaller than the required power. Dividing by m proves the first bound. Applying Stirling at $\mathbf{k}_N = N\mathbf{p}$ proves its asymptotic form.

Finally, with $L = |\ell|$, the ratio in (15.3) is exactly

$$\frac{(N-L-1)!}{(N-1)!} \prod_i \frac{(Np_i)!}{(Np_i - \ell_i)!} p_i^{-\ell_i},$$

which tends to one. Fixed shifts change none of the preceding power bounds along the ray. $\square$

15.2 One large summand

For sequences on $\mathbb{N}^q$, let $*$ denote additive convolution.

Theorem 15.2 (Directional one-large-summand principle). *For each fixed $m \geq 1$,*

$$\frac{(b^{*m})_{\mathbf{k}_N}}{b_{\mathbf{k}_N}} \longrightarrow m\mathfrak{B}^{m-1}. \quad (15.4)$$

Uniformly in $m \geq 1$ and all sufficiently large admissible N ,

$$\frac{(b^{*m})_{\mathbf{k}_N}}{b_{\mathbf{k}_N}} \leq C_0 m^{\varsigma+1} \mathfrak{B}^{m-1}. \quad (15.5)$$

Both assertions hold for a target with a fixed admissible shift.

Proof. In a decomposition $\mathbf{k}_N = \mathbf{k}^{(1)} + \cdots + \mathbf{k}^{(m)}$, some component has length at least N/m . Designate one such component. By Lemma 15.1 its weight is at most $C(N/m)^{-\varsigma}$. Summing the other $m-1$ components freely contributes at most $\mathfrak{B}^{m-1}$, and there are at most m choices for the designated index. The lower bound $b_{\mathbf{k}_N} \geq cN^{-\varsigma}$ proves (15.5). If $m > N$ the convolution vanishes, so no small-length exception is needed.

Now fix m and a cutoff H . Consider decompositions in which all but one component have length at most H . For $N > mH$ the large component is unique. The fixed-shift ratio (15.3) gives, after division by $b_{\mathbf{k}_N}$, the limit

$$m \left(\sum_{|\mathbf{k}| \leq H} b_{\mathbf{k}} \right)^{m-1}.$$

Every omitted decomposition has at least two components of length greater than H . Designate a largest one and one other such component. The same bound as before shows that their total normalized contribution is at most

$$C_m \mathfrak{B}^{m-2} \sum_{|\mathbf{k}| > H} b_{\mathbf{k}}$$

for $m \geq 2$. This tends to zero as $H \rightarrow \infty$. Taking first $N \rightarrow \infty$ and then $H \rightarrow \infty$ proves (15.4); $m = 1$ is immediate. The proof uses only finite total mass, not finite first moment. The shifted version follows from the shifted part of Lemma 15.1. $\square$

Corollary 15.3 (Entire functions of the bridge series). *For any fixed $z \in \mathbb{C}$,*

$$\frac{[\mathbf{x}^{\mathbf{k}_N}] \exp(z\mathcal{L}(\mathbf{p}\mathbf{x}))}{b_{\mathbf{k}_N}} \longrightarrow ze^{z\mathfrak{B}}. \quad (15.6)$$

In particular,

$$[\mathbf{x}^{\mathbf{k}_N}] \mathcal{E}(\mathbf{p}\mathbf{x}) \sim e^{\mathfrak{B}} b_{\mathbf{k}_N}.$$

Proof. For $N > 0$, expand the exponential and use (15.4) termwise. The absolute bounds

$$C_0 m^{\varsigma+1} \mathfrak{B}^{m-1} |z|^m / m!$$

are summable, so dominated convergence applies, also for negative or complex z . The limiting series is $ze^{z\mathfrak{B}}$. Tonelli's theorem applied to the nonnegative coefficient expansion also gives $\mathcal{E}(\mathbf{p}) = e^{\mathfrak{B}} < \infty$. $\square$

16 The general balanced-composition formula

Theorem 16.1 (Arbitrary rational balanced compositions). *Under (15.1), let $\kappa(\mathbf{p}) = \mathcal{E}(\mathbf{p}) = e^{\mathfrak{B}}$. Then*

$$e_{N\mathbf{p}} \sim \frac{\kappa(\mathbf{p})}{N} \binom{N}{Np_1, \dots, Np_q} \quad (16.1)$$

as $N \rightarrow \infty$ through admissible multiples. The constant $\kappa(\mathbf{p})$ is finite, positive, and algebraic.

Proof. Corollary 15.3 states $e_{N\mathbf{p}}\mathbf{p}^{N\mathbf{p}} \sim \kappa(\mathbf{p})b_{N\mathbf{p}}^0\mathbf{p}^{N\mathbf{p}}$. Cancel the common weight. Finiteness has already been proved. Algebraicity follows from the small-root formula at the rational critical weights, taking the finite limit from $t < 1$; equivalently take the product of its limiting algebraic roots. In particular, this argument does not need to infer algebraicity merely from convergence of a transcendental-looking exponential series. $\square$

For $p_i = 1/r$, $N = rn$, and the steps (3.1), Theorem 16.1 proves (3.5): this is source 42's proof of the leading term of Theorem 3.1, which uses no saddle point. Stirling gives

$$\frac{(rn)!}{(n!)^r} = \frac{\sqrt{r} r^{rn}}{(2\pi n)^{(r-1)/2}} \left(1 + \frac{r^{-1} - r}{12n} + O_r(n^{-2}) \right), \quad (16.2)$$

which proves the formula for $c(r)$ in (3.4) and hence the OEIS growth conjecture (2.2).

Theorem 16.1 and Corollary 12.1 meet on the step set (3.1) with centred rational probabilities, where they give the same leading term. Beyond that they are complementary: Theorem 16.1 covers every finite labelled list of integer steps with both signs and needs only the finite bridge mass, but gives the leading term only; Corollary 12.1 adds all orders and D-finiteness on its rays, for the rows' step set.

Remark 16.2 (A probabilistic interpretation). Among all balanced words, the probability of satisfying the prefix constraint is asymptotic to $\kappa_r/(rn)$. Thus κ_r measures the lattice correction to the naive inverse-length ballot probability. For $r = 2$ the exact probability is $1/(n+1)$; for $r = 3$ it is also $1/(n+1)$. The values $\kappa_2 = 2$ and $\kappa_3 = 3$ are consistent with both facts. There is no assertion that the analogous probability is exactly a rational function of n for larger r .

17 A second route to the complete expansion: lattice coordinates

Source 42 proves the all-orders part of Theorem 3.1 by the same strategy as Section 6 (Fourier extraction over the composition torus, a moving square-root singularity, coefficient transfer and Wick contraction), but in coordinates on the lattice of balanced compositions rather than marker coordinates, and with an explicit coefficient algorithm. Its formulation is printed here because Section 18 builds on it. Like Section 6, it uses local complex analysis, and the proof also specifies a finite coefficient procedure at every order.

17.1 Composition coordinates and the unique Fourier saddle

Fix r . Let

$$\Lambda = \{v \in \mathbb{Z}^r : \mathbf{1} \cdot v = \mathbf{s} \cdot v = 0\}, \quad d = r - 2.$$

Choose an integral basis of this lattice, with basis matrix $V \in \mathbb{Z}^{r \times d}$, and put $\Xi = (V^T V)^{-1} V^T$. Then

$$\Xi V = I_d, \quad \Xi \mathbf{1} = \Xi \mathbf{s} = 0. \quad (17.1)$$

Write $\xi_i \in \mathbb{Q}^d$ for the i th column of Ξ . At length $N = rn$, a zero-sum composition is uniquely $n\mathbf{1} + V\ell$ with $\ell \in \mathbb{Z}^d$, and $\Xi \mathbf{k} = \ell$. For $z \in \mathbb{C}^d$ set

$$P_z(u) = \frac{1}{r} \sum_{i=1}^r e^{\xi_i \cdot z} u^{s_i}, \quad \widehat{\mathcal{E}}(t, z) = \mathcal{E}\left(\frac{t}{r} e^{\xi_1 \cdot z}, \dots, \frac{t}{r} e^{\xi_r \cdot z}\right). \quad (17.2)$$

Fourier inversion, at precisely these lengths, gives

$$r^{-N} A_r(n) = \frac{1}{(2\pi)^d} \int_{[-\pi, \pi]^d} [t^N] \widehat{\mathcal{E}}(t, i\theta) d\theta. \quad (17.3)$$

For $d = 0$ the integral is absent.

The only saddle on this torus is zero, modulo its identified boundary. To see the needed strictness, suppose for some θ, ϕ that $|P_{i\theta}(e^{i\phi})| = 1$. Equality in the triangle inequality implies $\xi_i \cdot \theta + s_i \phi \equiv \gamma \pmod{2\pi}$ for all i . Pair this relation with any $v \in \Lambda$. It gives $\theta \cdot \Xi v \equiv 0$, so $\theta \in 2\pi\mathbb{Z}^d$. Hence outside a neighborhood of zero the maximum over ϕ is bounded strictly below one. The constant-term logarithm in (14.3) then converges uniformly for $|t| < 1 + \epsilon$, giving an exponentially small contribution to (17.3) there.

17.2 The critical point and the square-root expansion

There is a unique analytic function $\tau(z)$ near zero, with $\tau(0) = 1$, satisfying

$$u \frac{\partial P_z}{\partial u} \Big|_{u=\tau(z)} = 0.$$

This follows from the implicit-function theorem since

$$V = \frac{1}{r} \sum_i s_i^2 > 0.$$

Put $P_*(z) = P_z(\tau(z))$ and $\Psi(z) = \log P_*(z)$. The identities (17.1) imply

$$P_*(0) = 1, \quad \Psi'(0) = 0, \quad Q := \Psi''(0) = \frac{1}{r} \Xi \Xi^T > 0. \quad (17.4)$$

The notation Q in this Part is a positive matrix, not the kernel polynomial $Q_r(u)$, and not the Hessian H of Section 6, which is the corresponding matrix in marker coordinates.

At $t = P_*(z)^{-1}$, one small and one large root coalesce at $u = \tau(z)$. The second derivative in u is nonzero, so the analytic Morse lemma, or direct coefficient solving, gives a convergent Puiseux expansion in $(1 - tP_*(z))^{1/2}$. The product of the other small roots is analytic in t, z in this neighborhood. This remains true even if individual nonprincipal roots have an internal collision: the product of the roots inside a separating contour is analytic by symmetric root functions. Thus

$$\widehat{\mathcal{E}}(t, z) = \Upsilon_{\text{an}}(t, z) + \sum_{k \geq 0} \Upsilon_k(z) (1 - tP_*(z))^{k+1/2}, \quad (17.5)$$

locally, with analytic coefficients. Its leading coefficient is

$$\Upsilon_0(z) = -\kappa_*(z) \sqrt{\frac{2P_*(z)}{\tau(z)^2 P_z''(\tau(z))}}, \quad \kappa_*(z) = \widehat{\mathcal{E}}(P_*(z)^{-1}, z). \quad (17.6)$$

Here $\kappa_*(z)$ denotes the critical value on the continued small-root branch; it is not an assertion of absolute power-series convergence at every complex marked point. In particular $\Upsilon_0(0) = -\kappa_r \sqrt{2}/\sqrt{V}$.

The time period is $\nu = 1$ for odd r and $\nu = 2$ for even r . Indeed, odd r admits a zero step, while for even r all steps are odd and a two-step return is possible. At zero markings the only dominant singularities are the corresponding time rotations of one. This follows as well from the triangle-equality argument in the preceding subsection. At nearby complex markings their local contributions are the same on admissible lengths. Their separation from the remaining

singularities persists uniformly in a small neighborhood. Therefore coefficient transfer from (17.5) yields, for $N = rn$,

$$[t^N]\widehat{\mathcal{E}}(t, z) = \nu P_*(z)^N N^{-3/2} \left(\sum_{j=0}^M \frac{\mathcal{G}_j(z)}{N^j} + O(N^{-M-1}) \right), \quad (17.7)$$

uniformly locally in z , with

$$\mathcal{G}_0(z) = \frac{\kappa_*(z)}{\sqrt{2\pi}} \sqrt{\frac{P_*(z)}{\tau(z)^2 P''_z(\tau(z))}}. \quad (17.8)$$

Uniformity can be obtained by the same separated root contours and a common slit neighborhood for the coalescing pair. Outside those neighborhoods a coefficient contour can be pushed a fixed positive distance beyond the dominant radius. The analytic local terms have no singular contribution; the half-integral terms transfer by the gamma-ratio formula in the next subsection. These observations give the uniform remainder in (17.7), rather than merely a pointwise formal expansion.

17.3 An all-orders coefficient algorithm

For nonintegral γ define rational polynomials $t_i(\gamma)$ by

$$[t^N](1-t)^\gamma = \frac{N^{-\gamma-1}}{\Gamma(-\gamma)} \left(\sum_{i=0}^M \frac{t_i(\gamma)}{N^i} + O(N^{-M-1}) \right). \quad (17.9)$$

They are computed by expanding $\Gamma(N-\gamma)/\Gamma(N+1)$, as in (6.11); in particular $t_0 = 1$ and $t_1(\gamma) = \gamma(\gamma+1)/2$. Then

$$\mathcal{G}_j(z) = \sum_{k=0}^j \frac{\Upsilon_k(z)}{\Gamma(-k-1/2)} t_{j-k}(k+1/2). \quad (17.10)$$

Write $\Psi(z) = \frac{1}{2}z^\top Qz + \sum_{m \geq 3} \Psi_m(z)$, where Ψ_m is homogeneous of degree m . Let X be a centred d -dimensional Gaussian with covariance Q^{-1} . Form the formal series

$$\mathcal{A}(\hbar, X) = \left(\sum_{j \geq 0} \hbar^{2j} \mathcal{G}_j(i\hbar X) \right) \exp \left(\sum_{m \geq 3} \hbar^{m-2} \Psi_m(iX) \right). \quad (17.11)$$

Every coefficient needed below is a finite polynomial in X . Set

$$\chi_j = \mathbb{E}[\hbar^{2j}] \mathcal{A}(\hbar, X), \quad \alpha_{r,j} = \frac{\chi_j}{r^j \mathcal{G}_0(0)}. \quad (17.12)$$

Gaussian moments are computed exactly by pairings: an odd moment is zero, and an even moment is the sum, over pairings of its factors, of products of covariance entries. Thus (17.12) is a finite procedure, not an appeal to fitted coefficients.

Second proof of the all-orders part of Theorem 3.1 (source 42). Insert (17.7) into (17.3) and scale $\theta = X/\sqrt{N}$ in a neighborhood of the unique saddle. By (17.4),

$$N\Psi(iX/\sqrt{N}) = -\frac{1}{2}X^\top QX + \sum_{m \geq 3} N^{-(m-2)/2} \Psi_m(iX).$$

Taylor expansion of the analytic amplitudes gives exactly (17.11) with $\hbar = N^{-1/2}$. Terms of odd degree in $\hbar$ integrate to zero: in each such term the total degree in X is odd. The resulting expansion is

$$r^{-N} A_r(n) = \frac{\nu N^{-(d+3)/2}}{(2\pi)^{d/2} \sqrt{\det Q}} \left(\sum_{j=0}^M \chi_j N^{-j} + O(N^{-M-1}) \right).$$

To justify truncation, first restrict to $|X| \leq N^{\varepsilon_1}$ with sufficiently small fixed $\varepsilon_1 > 0$. Analytic Taylor remainders there are bounded by a chosen power of $N^{-1/2}$ times a polynomial in $|X|$ and a Gaussian envelope. Expand to enough extra orders to reach the prescribed M , and integrate that envelope. The part with $N^{\varepsilon_1} < |X|$ inside the fixed saddle neighborhood is exponentially small in $N^{2\varepsilon_1}$, since $\operatorname{Re} \Psi(i\theta) \leq -c|\theta|^2$. The complement of the saddle neighborhood was already exponentially controlled. This proves the stated remainder at every order.

The leading coefficient agrees with (3.5), already proved independently (Theorem 16.1); this also checks all lattice and time-period factors. Since $N = rn$ and $d + 3 = r + 1$, normalizing by the leading coefficient gives (3.3) and (17.12).

Finally, all local Taylor and Puiseux coefficients at zero are obtained by algebraic operations at algebraic roots of rational polynomials. Gamma values at the negative half-integers contribute only a common $1/\sqrt{\pi}$ times rational factors. The covariance is rational. Consequently $\chi_j/\mathcal{G}_0(0)$, and hence every $\alpha_{r,j}$, is algebraic. The finite steps just described establish effectiveness. $\square$

The scope of “complete” is the same as in the remark at the end of Section 6.

18 First corrections by Gaussian moments

18.1 A coordinate-invariant first-correction formula

The local calculation can use any rational full-rank $d \times r$ matrix Ξ annihilating $\mathbf{1}$ and $\mathbf{s}$, because an invertible linear change of local coordinates cancels from the normalized coefficients. In particular, consecutive second-difference rows $(1, -2, 1)$ are convenient. They need not be the globally normalized Fourier coordinates of Section 17; only the local ratios are being computed.

Put $L_i(x) = \xi_i \cdot x$ and define

$$\mathcal{M}_j(x) = \frac{1}{r} \sum_i L_i(x)^j, \quad \mathcal{M}_{12}(x) = \frac{1}{r} \sum_i s_i L_i(x)^2.$$

The critical phase has cubic and quartic terms

$$\Psi_3(x) = \frac{\mathcal{M}_3(x)}{6}, \quad \Psi_4(x) = \frac{\mathcal{M}_4(x) - 3\mathcal{M}_2(x)^2}{24} - \frac{\mathcal{M}_{12}(x)^2}{8V}. \quad (18.1)$$

Indeed, writing $v = \log \tau(z)$, the critical equation for $\log(r^{-1} \sum_i e^{\xi_i \cdot z + s_i v})$ gives $v = -\mathcal{M}_{12}(z)/(2V) + O(|z|^3)$. Substitution into its cumulant expansion proves (18.1).

Let $\mathbf{a}(z) = \mathcal{G}_0(z)/\mathcal{G}_0(0)$ and write $\mathbf{a}' = \nabla \mathbf{a}(0)$ and $\mathbf{a}'' = \nabla^2 \mathbf{a}(0)$. Equations (17.11)–(17.12) give

$$\begin{aligned} r \alpha_{r,1} &= \frac{\mathcal{G}_1(0)}{\mathcal{G}_0(0)} - \frac{1}{2} \operatorname{tr}(Q^{-1} \mathbf{a}'') \\ &\quad + \mathbb{E} \left[\frac{(\mathbf{a}' \cdot X) \mathcal{M}_3(X)}{6} + \Psi_4(X) - \frac{\mathcal{M}_3(X)^2}{72} \right]. \end{aligned} \quad (18.2)$$

The negative sign in the final term comes from $i^6 = -1$. Omitting it changes the answer.

18.2 Computing the local kernel terms

Here are explicit formulas sufficient to reproduce the first correction. At zero markings put $P = P_0$ and expand the principal small root at $t = 1 - w^2$ as $u = 1 + a_1 w + a_2 w^2 + a_3 w^3 + O(w^4)$, as in Appendix A. Solving $P(u) = 1/(1 - w^2)$ gives

$$\begin{aligned} a_1 &= -\sqrt{2/P''(1)}, & a_2 &= -\frac{P'''(1)a_1^2}{6P''(1)}, \\ a_3 &= \frac{1 - \frac{1}{2}P''(1)a_2^2 - \frac{1}{2}P'''(1)a_1^2a_2 - \frac{1}{24}P''''(1)a_1^4}{P''(1)a_1}. \end{aligned} \quad (18.3)$$

When the nonprincipal interior roots $\omega \in \mathcal{R}_r$ are simple, where $\mathcal{R}_r$ is the multiset of roots of $Q_r(u)/(u-1)^2$ in $|u| < 1$, their w^2 coefficients are $1/P'(\omega)$. Therefore

$$\frac{\mathcal{G}_1(0)}{\mathcal{G}_0(0)} = \frac{3}{8} - \frac{3}{2} \left(\frac{a_3}{a_1} + 1 + \sum_{\omega \in \mathcal{R}_r} \frac{1}{\omega P'(\omega)} \right). \quad (18.4)$$

For nonsimple nonprincipal roots, compute the derivative of their analytic product instead of differentiating the individual roots; the general algorithm in Section 17 is unaffected. The roots for $r = 3, 5$ used below are simple (or absent).

For the amplitude derivatives, use

$$\log \mathcal{G}_0(z) = \log \kappa_*(z) - \frac{1}{2} \log V(z) - \frac{1}{2} \log(2\pi), \quad V(z) = \left. \frac{\partial^2}{\partial v^2} \log P_z(e^v) \right|_{v=\log \tau(z)}.$$

The root product gives

$$\log \kappa_*(z) = \log P_*(z) + \log \tau(z) + \sum_{\omega} \log \omega(z) - \xi_- \cdot z + \text{constant}.$$

Here ξ_- is the column for the minimum step. The derivatives of a simple nonprincipal root follow by implicit differentiation of $P_z(\omega(z)) = P_*(z)$:

$$\begin{aligned} \omega_j &= -P_{z_j}(\omega)/P'(\omega), \\ (\log \omega)_{jk} &= \frac{Q_{jk} - P_{z_j z_k}(\omega) - P_{uz_j}(\omega)\omega_k - P_{uz_k}(\omega)\omega_j - P''(\omega)\omega_j\omega_k}{\omega P'(\omega)} - \frac{\omega_j\omega_k}{\omega^2}. \end{aligned} \quad (18.5)$$

All quantities on the right are evaluated at zero markings. In addition,

$$(\log \tau)_{jk} = -\frac{1}{rV} \sum_i s_i \xi_{i,j} \xi_{i,k}, \quad V_j = \frac{1}{r} \sum_i s_i^2 \xi_{i,j}, \quad V_{jk} = \frac{1}{r} \sum_i s_i^2 \xi_{i,j} \xi_{i,k} - V Q_{jk}.$$

The last identity uses the symmetry $\sum_i s_i^3 = 0$. These formulas make (18.2) an exact algebraic calculation.

18.3 Three evaluated cases

The two exact rows of Section 7 give the second-order terms

$$\begin{aligned} A_2(n) &= \frac{4^n}{\sqrt{\pi} n^{3/2}} \left(1 - \frac{9}{8n} + \frac{145}{128n^2} + O(n^{-3}) \right), \\ A_3(n) &= \frac{\sqrt{3} 27^n}{2\pi n^2} \left(1 - \frac{11}{9n} + \frac{101}{81n^2} + O(n^{-3}) \right). \end{aligned} \quad (18.6)$$

For $r = 5$, formulas (18.2)–(18.5) give the value of $\alpha_{5,1}$ in (3.6),

$$\alpha_{5,1} = -\frac{13}{10} + \frac{13\sqrt{5}}{50},$$

by a contraction different from the one of Section 9. In the multinomial normalization this is

$$\frac{A_5(n)}{\kappa_5(5n)!/(5n(n!)^5)} = 1 + \frac{-9/10 + 13\sqrt{5}/50}{n} + O(n^{-2}), \quad (18.7)$$

that is, $\Delta_5/5$ in (9.3). The shipped `code/42-late-growing-first_correction.py` carries out the rational Gaussian moments and algebraic root differentiation exactly for $r = 3, 5$. It obtains

$$\alpha_{3,1} = -11/9, \quad \alpha_{5,1} = -13/10 + 13\sqrt{5}/50 = -0.71862232585005467893\dots$$

The value of $\alpha_{3,1}$ is obtained here from the general formula and agrees with the value read off the closed form in Section 7. These values are not estimates from sequence data and do not use the conjectured recurrence.

19 Primitive excursions and the giant component

An excursion is *primitive* if it is nonempty and has no zero-height prefix before its endpoint; primitive excursions are the arches of Section 10. A one-step zero excursion is primitive. Every excursion factors uniquely at its returns to height zero. Let $\mathcal{I}(\mathbf{x})$ enumerate primitive excursions; then

$$\mathcal{E} = \frac{1}{1 - \mathcal{I}}, \quad \mathcal{I} = 1 - e^{-\mathcal{L}}. \quad (19.1)$$

We retain the general critical composition (15.1) in the first theorem and put $\kappa = \mathcal{E}(\mathbf{p})$.

Lemma 19.1 (Primitive weights). *If $i_{\mathbf{k}} = [\mathbf{x}^{\mathbf{k}}]\mathcal{I}(\mathbf{p}\mathbf{x})$, then*

$$0 \leq i_{\mathbf{k}} \leq b_{\mathbf{k}}, \quad \sum_{\mathbf{k}} i_{\mathbf{k}} = 1 - \kappa^{-1}, \quad i_{\mathbf{k}_N} \sim \kappa^{-1} b_{\mathbf{k}_N}.$$

The last assertion holds also after a fixed admissible shift.

Proof. A primitive excursion of length $m > 1$ has a unique minimum among its cyclic vertices before the endpoint. Its m rotations are distinct: a nontrivial word period would give an earlier zero return. No rotation class can contain two primitive excursions. Consequently m times the number of primitive words of a fixed composition is at most the number of all bridge words of that composition, proving $i_{\mathbf{k}} \leq b_{\mathbf{k}}$. Length one is immediate. The mass identity follows from (19.1). Finally apply Corollary 15.3 with $z = -1$ and use $\mathcal{I} = 1 - e^{-\mathcal{L}}$. $\square$

Theorem 19.2 (A size-biased geometric law). *Choose uniformly an excursion of composition $N\mathbf{p}$, and let J_N be its number of primitive factors. For each fixed $k \geq 1$,*

$$\mathbb{P}(J_N = k) \longrightarrow k\kappa^{-2}(1 - \kappa^{-1})^{k-1}. \quad (19.2)$$

All fixed moments converge. In particular,

$$\mathbb{E}J_N \longrightarrow 2\kappa - 1, \quad \mathbb{P}(J_N = 1) \longrightarrow \kappa^{-2}. \quad (19.3)$$

For the OEIS rows take $\kappa = \kappa_r$ and $N = rn$.

For the rows this is the law of Theorem 10.1, proved there by the analytic transfer of Section 6: J_N is the number J_n of arches, $C_n = J_n - 1$, and $2\kappa_r - 1 = 1 + 2(\kappa_r - 1)$. The proof below is source 42's second route, by the cutoff argument of Theorem 15.2; it also covers every centred rational composition and every fixed moment.

Proof. Repeat the cutoff proof of Theorem 15.2 for i in place of b . Lemma 19.1 supplies the same global bound and the local ratio $i_{\mathbf{k}_N - \ell} / b_{\mathbf{k}_N} \rightarrow \kappa^{-1}$. With $\eta = 1 - \kappa^{-1}$, it follows that

$$(i^{*k})_{\mathbf{k}_N} / b_{\mathbf{k}_N} \longrightarrow k \kappa^{-1} \eta^{k-1}.$$

Divide by the total excursion coefficient, asymptotic to $\kappa b_{\mathbf{k}_N}$. A uniform bound of the form $C k^{\varsigma+1} \eta^{k-1}$ follows by designating a largest component. Since $\eta < 1$, it proves convergence of all fixed moments by dominated summation. The limiting probability generating function is

$$\frac{v}{\kappa^2(1 - \eta v)^2}. \quad (19.4)$$

Differentiating at $v = 1$ proves the mean formula. $\square$

Theorem 19.3 (One giant primitive component). *Write a uniform excursion of composition $N\mathbf{p}$ as $\mathcal{U}_N \Omega_N \mathcal{V}_N$, where Ω_N is a longest primitive component and ties are resolved by choosing the first one. Then $(\mathcal{U}_N, \mathcal{V}_N)$ converges in total variation to a pair $(\mathcal{U}, \mathcal{V})$ of independent finite excursions with*

$$\mathbb{P}(\mathcal{U} = u) = \frac{\mathbf{p}^{\text{comp}(u)}}{\kappa}, \quad \mathbb{P}(\mathcal{V} = v) = \frac{\mathbf{p}^{\text{comp}(v)}}{\kappa}. \quad (19.5)$$

In particular $N - |\Omega_N| = O_{\mathbb{P}}(1)$.

Proof. Fix two concrete finite excursions u, v and put $\ell = \text{comp}(u) + \text{comp}(v)$. For all sufficiently large N , an excursion of the form ujv with j primitive of composition $N\mathbf{p} - \ell$ automatically has j as its unique longest factor. Its probability is the ratio of that primitive count to $e_{N\mathbf{p}}$. The shifted primitive asymptotic gives the limit $\mathbf{p}^\ell / \kappa^2$, exactly the product in (19.5).

The sum of these limiting probabilities over all ordered pairs of finite excursions is $\mathcal{E}(\mathbf{p})^2 / \kappa^2 = 1$. Pointwise convergence on this countable space to a distribution of total mass one implies total-variation convergence: restrict first to a finite subset with limiting mass at least $1 - \epsilon$, then control its complement. Tightness of the two finite decoration lengths follows. It also shows that the probability of competing macroscopic factors tends to zero. $\square$

This limit is compatible with a nontrivial number of small factors: a giant *primitive* component need not mean that the whole excursion is primitive.

19.1 A tight gap with an infinite-mean limit

For the uniform alphabet, let $D_{\text{dec}} = |\mathcal{U}| + |\mathcal{V}|$ be the limiting decoration length. Its probability generating function is

$$\mathbb{E} t^{D_{\text{dec}}} = \left(\frac{\mathcal{E}(t/r, \dots, t/r)}{\kappa_r} \right)^2. \quad (19.6)$$

The square-root expansion at $t = 1$ gives

$$\mathbb{E} t^{D_{\text{dec}}} = 1 - \frac{2\sqrt{2}}{\sqrt{V}} \sqrt{1-t} + O(1-t).$$

Transfer and summation along the permitted time lattice yield

$$\mathbb{P}(D_{\text{dec}} \geq m) \sim \frac{2\sqrt{2}}{\sqrt{V}\pi} m^{-1/2}, \quad \mathbb{E}D_{\text{dec}} = \infty. \quad (19.7)$$

For even r , the support is even; the factor two in coefficient transfer cancels the half-density when the tail is summed. Thus (19.7) holds as written for unrestricted integer $m \rightarrow \infty$. Tightness of $N - |\Omega_N|$ does not imply convergence of its expectation to a finite constant. This is a useful warning against an otherwise tempting interpretation of the giant-component theorem.

20 Ratios, curvature and an explicit rounding bound

20.1 Ratios and eventual strict log-convexity

With $\beta = (r+1)/2$ and $\lambda = r \log r$ as in Section 11, the complete expansion gives

$$\frac{A_r(n+1)}{A_r(n)} = r^r \left(1 - \frac{\beta}{n} + \frac{\beta(\beta+1)/2 - \alpha_{r,1}}{n^2} + O_r(n^{-3}) \right). \quad (20.1)$$

Taking sufficiently many terms before taking second differences gives

$$\log A_r(n+1) - 2 \log A_r(n) + \log A_r(n-1) = \frac{\beta}{n^2} + O_r(n^{-3}) > 0 \quad (20.2)$$

for all sufficiently large n . Therefore every fixed row is eventually strictly log-convex, and its successive ratios eventually increase to r^r from below. This is not a claim of log-convexity at every finite index. A remainder controlled only by $o(1)$ would not suffice to infer the curvature; the higher-order expansion supplies the needed control.

20.2 The inverse dominant sector

Source 42 inverts the dominant sector exactly as Section 11 does: the large real solution of $Y = K_r e^{\lambda x} x^{-\beta}$ is (11.2), on the lower real branch (the principal branch would select the small solution, not the large inverse); the formula follows by raising the model equation to the power $-1/\beta$ and applying the defining equation $W e^W = z$. Its correction terms are those of (11.4), with $b_1 = \alpha_{r,1}$ and $b_2 = \alpha_{r,2} - \frac{1}{2}\alpha_{r,1}^2$, obtained by successive formal substitution or Newton correction; at every order the next coefficient is determined linearly, since the derivative of the leading logarithm is $\lambda - \beta/x$, bounded away from zero for large x . For orientation, writing $L_Y = \log Y$ gives

$$x(Y) = \frac{L_Y + \beta \log L_Y - \log K_r - \beta \log \lambda}{\lambda} + O_r \left(\frac{\log L_Y}{L_Y} \right). \quad (20.3)$$

For A215570 the first correction to the Lambert model is positive, because $\alpha_{5,1} < 0$; compare (11.5).

20.3 Discrete thresholds and the rounding boundary

Let $N_r(Y) = \min\{n : A_r(n) \geq Y\}$ for sufficiently large Y , as in Section 11.3. The sequence is eventually strictly increasing by (20.1), so this tail threshold is unambiguous. Let $x_M(Y)$ solve the logarithmic expansion truncated after x^{-M} . There exists $\varpi_{r,M}$ such that, for all large Y ,

$$\left\lceil x_M(Y) - \varpi_{r,M}(\log Y)^{-M-1} \right\rceil \leq N_r(Y) \leq \left\lceil x_M(Y) + \varpi_{r,M}(\log Y)^{-M-1} \right\rceil. \quad (20.4)$$

This follows by comparing the logarithmic error at integer arguments with the derivative bounded away from zero. Away from the shrinking neighborhoods of integers the two bounds coincide. Near an integer, the error interval is necessary; a blanket claim that one may always take a ceiling of a truncated transseries would be false.

Remark 20.1 (Repository context). Inequality (20.4) is an instance of parts (2)–(3) of the staircase theorem `p0:thm:staircase` of the canonical transseries volume (Section 1.4), with $\eta = \varpi_{r,M}(\log Y)^{-M-1}$; its separation clause is formalized as the Lean theorem `Fabius.staircase_separation` and the failure of separation for arbitrarily small η as `Fabius.staircase_separation_fails`. Those are general statements about ceilings; they do not verify the asymptotic inputs of this report, and no novelty is claimed for the rounding statement.

21 Source 42’s computations

21.1 Exact dynamic programming

For an arbitrary composition $\mathbf{k}$, let $\text{ht}(\mathbf{k}) = \mathbf{s} \cdot \mathbf{k}$. Excursion-prefix counts satisfy

$$F(\mathbf{0}) = 1, \quad F(\mathbf{k}) = \begin{cases} 0, & \text{ht}(\mathbf{k}) < 0, \\ \sum_{i:k_i>0} F(\mathbf{k} - \mathbf{e}_i), & \text{ht}(\mathbf{k}) \geq 0. \end{cases} \quad (21.1)$$

The desired value is $F(n\mathbf{1})$; this is the recurrence of Section 13, implemented independently. This recursion is exact: every admissible word is obtained uniquely by deleting its last letter. The implementation traverses a rectangular composition box in lexicographic order, which is topological for coordinate decrements. It uses integer arithmetic and no row recurrence conjecture.

Two companion arrays count primitive words and the sum of the numbers of returns. For the primitive array, do not propagate any nonempty zero-height state. For the moment array, add the current count whenever a newly reached state has height zero. This gives independent finite tests of both (19.2) and (19.3).

r	n	$A_r(n)/(\kappa_r(rn)!/(rn(n!)^r))$	Primitive probability	Mean returns
4	30	0.993523253212	0.527972597714	1.74565492347
5	15	0.979218847813	0.283110782306	2.70465628143
6	9	0.991084348462	0.662077457976	1.45242243648
7	5	0.971363174805	0.417632780048	2.04381815151

Table 4: Exact integer counts of source 42 evaluated against the proved limiting formulas. These modest-index ratios illustrate convergence but do not certify asymptotic error bounds. “Mean returns” is the mean number $\mathbb{E}J_N$ of primitive factors, whose limit is $2\kappa_r - 1$ (Table 3).

21.2 Independent coefficient and algebra checks

Source 42’s program performs the following finite checks.

- (a) The first seven $r = 4, 5$ values are checked against the displayed OEIS data. Full independently computed ranges are $0 \leq n \leq 30$ for $r = 4$, $0 \leq n \leq 15$ for $r = 5$, $0 \leq n \leq 9$ for $r = 6$, and $0 \leq n \leq 5$ for $r = 7$.

- (b) The bridge–excursion identity is checked coefficientwise in small multivariate boxes using the independent Euler identity (13.1). The tested boxes use $(r, \text{bound}) = (2, 5), (3, 4), (4, 3), (5, 2)$.
- (c) The sextic elimination identity is verified as an exact rational-function identity, including its specified branch at the origin.
- (d) Kernel polynomials are deflated exactly by $(u - 1)^2$. Their interior roots are evaluated at high precision for $2 \leq r \leq 12$ and the four radical constants are cross-checked. These decimal evaluations are not interval-certified root isolations.
- (e) The first-correction formula is evaluated symbolically for $r = 3, 5$ using exact Gaussian moments. The $r = 3$ result is independently checked against its elementary exact formula.

The shipped `data/42-late-growing-verification.txt` and `data/42-late-growing-first_correction.txt` record the successful runs. The CSV files retain exact counts and decimal diagnostics. All the supplied assertions passed in the recorded run. The full dynamic program stores arrays of size $(n + 1)^r$, so memory, rather than asymptotic arithmetic sophistication, is the main practical limitation of this deliberately simple independent verifier.

21.3 Claims that remain outside the certificate

The proofs are mathematical arguments in this report, not Lean or Rocq proofs. The scripts certify their stated finite computations, not every analytic estimate. No numerical fit is used to prove an asymptotic coefficient. No low-order recurrence is used as an assumption. No actual recurrence operator for $r = 4$ is supplied. No claim of exhaustive literature priority, uniformity in r , exponentially improved asymptotics, or an explicit finite log-convexity threshold is made.

Part III

Research directions and conclusion

22 Further research questions and topics

The research directions of the two sources are merged here: each subsection states source 61's question and then what source 42 adds; two subsections are source 42's alone. The eight questions of source 75-03 (Part IV) were added on October 2, 2026, as sentences marked with its name. The A215570 question of sources 61 and 42 is answered by source 75-03, and source 75-03's question on all orders for every row by Theorem 3.1.

22.1 Certify the explicit A215570 recurrence (resolved)

The sextic (4.5), its specified formal branch, and a diagonal representation give a concrete input for creative telescoping. The next objective is to compute a certified annihilator and prove that the order-three operator in Kauers–Koutschan Conjecture 15 is a right factor annihilating the same solution, with all initial conditions and singular indices checked. Merely matching additional terms would not accomplish this. A useful outcome would be a compact certificate independently checkable without a large symbolic-computation environment. Source 42 (its project 1) proposes the same route and asks for a machine-readable telescoping certificate

and an independent checker: resolving the named conjecture, not just the existence of some recurrence, is the point.

Resolved (2026-10-02). Source 75-03 proves the conjecture (Theorem 26.1) with exactly such a compact certificate (Appendix C) and an exact checker, for every $n \geq 0$, so that no initial conditions or singular indices remain to be checked. Its route differs from the one proposed here: it does not eliminate from the sextic, but reduces the diagonal to the one-variable coefficient (25.7) and telescopes its Legendre residue representation. Minimality of the order-three operator is not claimed. The recurrence questions for A215562 and the higher rows remain open (next subsection).

22.2 Find and explain the fourth-row recurrence

A215562 is D-finite by the proof here, but its minimal recurrence and its computational complexity remain to be determined. The even-step parity and the larger kernel degree should be incorporated before elimination, not left for a generic telescoper to rediscover. What upper bounds on recurrence order and coefficient degree can be proved as functions of r ? Do the even and odd alphabets have qualitatively different minimal orders? Source 42 (project 2) notes that the generic small-root subset polynomial has degree at most $\binom{6}{3} = 20$ in this case, proposes exploiting the reciprocal and label-reversal symmetries before telescoping, and asks for a comparison of recurrence order, coefficient degree and bit complexity with direct composition dynamic programming. Source 75-03 asks for the same operator: eliminate the product of three small roots of the marked degree-six kernel and compute a diagonal telescoper, ideally with a compact rational certificate. The degree bound 20 suggests a larger problem than the five-symbol sextic but does not predict the minimal order; a smaller auxiliary coefficient representation, analogous to (25.7), would be especially valuable.

22.3 Automate all-order coefficient extraction

The proof supplies finite algebraic operations, but a practical system should exploit residual-marker coordinates, symmetry, and Wick contractions to avoid dense high-dimensional tensors. An immediate target is exact $\alpha_{r,1}, \alpha_{r,2}, \alpha_{r,3}$ for $r = 4, 6, 7, 8$, together with machine-readable algebraic-number specifications. For $r = 5$, the field $\mathbb{Q}(\sqrt{5})$ makes an especially clean regression suite. Source 42 (project 3) proposes the same engine built on (17.10) and (17.12), with exact algebraic numbers, branch-isolation certificates, symmetry reduction and regression tests from $r = 2, 3$, and targets $\alpha_{r,2}, \alpha_{r,3}$ for $r = 4, 5, 6$; its first-correction program is a prototype, not this engine. For $r = 5$ the target is met through $\alpha_{5,8}$ (Proposition 27.1), by the recurrence rather than by such an engine. Source 75-03 asks whether its transfer argument, which gives only a nonsharp $O(n^{-1/5})$ error, can be refined to all orders for every fixed alphabet; Theorem 3.1 already does so, so its question reduces to the targets above (it names an explicit, proved $1/n$ correction for $r = 4, 6, 7$) and to a uniform version of the convolution method giving a better general error without the whole expansion. It also asks for the minimal polynomials and degrees of the constants (5.6) and their dependence on parity, whether reciprocal symmetry reduces the generic subset-product degree bound substantially, and for rational isolating boxes of the inner roots that prove decimal intervals for κ_r .

22.4 Determine sharp asymptotic error constants

Turn the separated kernel contours and local Taylor bounds into explicit numerical inequalities. This would provide certified approximation intervals at finite n , certified threshold inversion near an integer boundary, and explicit eventual-log-convexity thresholds (source 42, project 4).

Source 75-03 asks the same for the constant C of its rounded threshold for A215570 (Section 28): explicit finite- Y bounds that bracket exact threshold searches.

22.5 Determine the exponentially small sectors

The dominant expansion does not identify subdominant critical points, connection constants, or Stokes phenomena. Once an explicit differential or recurrence operator is available, compare its formal solutions with complex kernel saddles. Which exponentially small contributions are actually present in the combinatorial solution? Can optimal truncation and rigorous exponentially improved error bounds be obtained without first computing a minimal recurrence? Source 42 (project 9) asks the same: identify non-dominant singularities of the algebraic kernel and the relevant complex composition saddles, which of them contribute after diagonal extraction, their exponential scales, and their Stokes transitions under parameter changes. This is what would be needed to replace the dominant-sector expansion by a more complete transseries. For A215570 the question is now concrete (source 75-03): the proved recurrence has normalized characteristic roots $1, 1, 2/27$ (Remark 27.2). Determine the decomposition of the actual solution into its dominant and exponentially smaller components, with connection constants, and whether and how optimally truncated dominant expansions detect the third mode.

22.6 Let the alphabet grow

Our estimates keep r fixed. When $r = r(n)$ grows, kernel-root separation, Gaussian dimension, and constants in the remainder estimates all change. Determine useful uniform ranges, and analyze the large- r behavior of κ_r separately along even and odd alphabets. This would connect the fixed-row problem to the less regular column direction of A215561. Source 42 (project 5) adds that the parity split is real: odd alphabets include a zero step, while even alphabets have time period two, so fixed- r error constants cannot simply be reused. It asks for expansions of $\log \kappa_r$ separately for even and odd r , and for a range of joint growth $r = r(n)$ in which the fixed-row formula remains uniform. Source 75-03 asks for the same simultaneous limit and, separately, for the column $n = 1$ of the array, which is A147681; none of the fixed-row theorems settles that column problem.

22.7 Resolve the mean size of the non-giant remainder

The limit gap D_{dec} has an $m^{-1/2}$ tail and infinite mean (19.7). Determine the growth of $\mathbb{E}(N - |\Omega_N|)$ and higher moments under exact composition conditioning, including their leading constants. Establish a transition law when the gap is a positive fraction of N and quantify the probability of two large components. Weak convergence alone does not answer these questions (source 42, project 6).

22.8 Approach the boundary of the composition simplex

In Corollary 12.1, positivity of all probabilities is essential for the full-dimensional saddle. If some p_i tend to zero, the determinant in (6.6) degenerates and the effective alphabet may change. Find uniform transition formulas interpolating between adjacent alphabet sizes and identify the scaling regimes in which a different asymptotic power appears. Source 42 (project 8) also asks at which scale the giant-component law of Theorem 19.3 changes; its Theorem 16.1 likewise assumes that every p_i is strictly positive. Source 75-03 asks the same when some type frequencies tend to zero with N : the multinomial covariance degenerates, the effective number

of types can change and the power $(r + 1)/2$ need not stay uniform; a useful theory would connect the full-dimensional regime to the lower-dimensional faces with explicit crossover scales.

22.9 Refine path statistics under exact composition

The contact law is only one consequence of the marked kernel. Natural next targets are its finite- n corrections, the maximum height, the area under the excursion, and functional convergence after diffusive scaling. These require additional arguments: neither the count asymptotic nor the contact law alone proves a Brownian-excursion limit or independence of other statistics. Source 42 (project 7) asks for the functional limit after diffusive height scaling jointly with the Boltzmann decorations of Theorem 19.3 and the primitive-factor count. Exact composition is a stronger conditioning than an endpoint constraint, and a satisfactory theorem should specify the covariance normalization and decide whether the limiting giant shape is asymptotically independent of the decorations. Source 75-03 asks for the same functional limit under exact content, expecting a Brownian-excursion-type limit with a content-determined variance, and names the height maximum, the area and the locations of short primitive components as observables; these need probability arguments beyond the scalar asymptotic.

22.10 Formalize the exact-to-analytic interface

A staged ProveIt development could first formalize the word–walk bijection, the finite-state recurrences, the bridge coefficient identity, and the formal algebraic kernel relation. A second stage would address algebraic-series diagonal closure and critical-root selection. The most substantial analytic stage is a reusable uniform marked square-root transfer theorem with a nondegenerate composition saddle and explicit lattice periods. Keeping these stages separate would prevent finite computations from being mistaken for proofs of analytic hypotheses. Source 42 (project 10) proposes to begin instead with its convolution route: the exact bridge coefficients, the factorial ratio lemma, the convolution cutoff bound and dominated exponential summation (Section 15) give the leading growth theorem with a relatively small analytic interface, so that the substantial all-orders complex analysis need not be a prerequisite for a formally verified leading result. The kernel identity and the multivariate saddle estimates can be later modules. Source 75-03 proposes a formalization of the recurrence proof as a chain of small reusable certificates: content vectors, prefix validity, reversal and zero-step insertion; the marked kernel algebra and the Lagrange extraction; the coefficient identities and the polynomial certificate in a rational-function ring; the Legendre recurrence and derivative identities with the vanishing residue of a rational derivative; and, separately, the Fourier bounds and the composition-transfer theorem. These are distinct obligations: checking the printed certificate alone does not establish the counting semantics.

23 Conclusion

The fixed-row problem is governed by a finite one-dimensional step set, not by an intrinsically growing transfer matrix. Marking composition before taking a diagonal makes this structure visible. The resulting algebraic ambient series gives D-finiteness; its bridge logarithm gives an exact reference coefficient; and a marked critical-point calculation converts that coefficient into the conjectured row asymptotic with an algebraic constant.

The same analysis yields a complete dominant inverse-power expansion, a concrete correction for A215570, a contact distribution, and an asymptotic inversion procedure. A separate

arithmetic argument shows that every row generating function above the Catalan case is transcendental. These results settle the stated fixed-row growth conjecture and the recurrence-existence issue while leaving the explicit recurrence certificates and exponentially small sectors as sharply specified next problems. (2026-10-02 note: Part IV supplies the certificate for A215570; the one for A215562 remains open.)

Source 42 reaches the leading formula by a different mechanism: along a strictly positive balanced composition ray, the coefficients of an exponential of the log-excursion series are dominated by one large summand, and the remaining summands contribute exactly the critical excursion value. The same mechanism, applied to the factorization into primitive excursions, explains the limiting number of returns and the single giant component with its heavy-tailed finite decorations. Certifying minimal recurrences, obtaining effective finite error bounds, and passing to growing alphabets are distinct next steps, not conclusions hidden inside the present theorems.

Source 75-03, finally, finds for the five-symbol row an exact representation much smaller than the content-state dynamic program: a positive one-variable coefficient with a Legendre-polynomial interpretation. It yields an explicit proof of the published order-three recurrence and, with the first correction, every dominant asymptotic coefficient and its inverse, while an explicit four-symbol recurrence, uniform growing-alphabet limits and the exponentially small sectors stay separate tasks.

Part IV

The Kauers–Koutschan recurrence and the five-symbol row

Part IV was added on October 2, 2026, in batch 75 of ProveIt’s incoming-reports intake, from a third manuscript on the same problem (source 75-03), after the closing Part III. It proves what Parts I and II left open and Part III lists as an open question: the explicit order-three recurrence for A215570 conjectured by Kauers and Koutschan (Theorem 26.1). With that recurrence it computes the exact coefficients $\alpha_{5,2}, \dots, \alpha_{5,8}$ of (3.3) for $r = 5$ (Table 6). About two thirds of source 75-03 re-proves results of Parts I and II, some of them in a weaker form; those results are *not* printed again but listed, with the comparison, in Table 5. The rest of the report is unchanged apart from dated notes that point here, rows added to Tables 1 and 2, the merged research questions of Part III and the package description of Appendix B; no existing label or number was changed.

24 Provenance, overlap and conventions of Part IV

24.1 A third manuscript

Source 75-03 is batch-75 manuscript 03, *Fixed-Content Ballot Words. A proof of the A215570 recurrence, a fixed-row asymptotic theorem, and all-orders inversion*, dated October 1, 2026 and prepared for Vladimir Reshetnikov (a 917-line source with 87 labels and a 22-page PDF). Archive `OEIS_Fixed_Content_Ballot_Words.zip`, arrival commit `4b874cea0`, placed in commit `6ea60e367`; it pins the repository at `29aca108e` [13]. Its programs and data ship with the prefix `75-03-ballot-words-` (Section 29).

Source 75-03 was written independently of sources 61 and 42: about 1.8% of its 8-word

shingles occur in this report (0.8 % of the report’s occur in it), all of them preamble, bibliography and OEIS strings. Its pin (October 1, 19:03) predates this report’s placement (6e193dd4f, 19:16) by thirteen minutes and already contained the archives of sources 61 and 42 as unopened zip files. Its repository check (“its overview, combinatorics overview, and a targeted search for A215561”) could therefore not see this report, exactly as source 61’s search could not see source 42 (Section 2.3). It states that the check “was not an audit of every research artifact”. Its own status statement is kept: its results “have explicit mathematical proofs; the large recurrence identity additionally has an exact symbolic checker. They have not been independently peer-reviewed or formalized in Lean”; its source check of October 1, 2026 found the statements proved here still recorded as conjectural, but “it is not an exhaustive priority search”; and classical kernel, diagonal and saddle-point methods are credited rather than claimed.

Like sources 61 and 42, source 75-03 is AI-assisted and unrefereed, and no Lean or Rocq development checks any statement of this Part. Source 75-03 outlines a staged formalization (Section 22, last subsection); none exists.

24.2 What the third manuscript re-proves

Table 5 lists every result of source 75-03 that Parts I and II already contain, where it is printed, and how the two versions compare. Its main fixed-row theorem is a *third* proof of the leading term of Theorem 3.1, by the one-large-summand route of source 42 (Section 15) made quantitative, and it is weaker than Theorem 3.1: a relative error $O(n^{-1/5})$ instead of a complete expansion. Source 75-03 presents it as an independent proof of the A215561 conjecture; in this report it is the third.

Table 5: Results of source 75-03 that Parts I and II already prove.
“Same” means the same statement; the last column records where source 75-03 is weaker, stronger or merely different.

source 75-03	printed here	comparison
bridge factorization $\mathcal{E} = \exp \mathcal{L}$, algebraicity, small-root formula (its Thm. 2.1)	Lemmas 4.1, 5.1; (14.3)	same; the residue proof is source 61’s first proof
D-finiteness of every row (Cor. 2.2)	Theorem 3.2	same; its degree bounds $\binom{r-1}{(r-1)/2}$ (r odd) and $\binom{2r-2}{r-1}$ (r even), 6 for $r = 5$ and 20 for $r = 4$, are the bound $\binom{h+h'}{h}$ of Section 14
mass, central size, fixed shifts (Lemma 3.1)	Lemma 15.1	same, plus the uniform shift rate $1 + O((J + J^2)/N)$ and the tail bound $O(M^{-1/2})$ (Remark 24.1)
central composition transfer (Thm. 3.2), balanced excursions (Cor. 3.3)	Theorem 15.2, Corollary 15.3, Theorem 16.1	adds the rate $O(N^{-1/5})$ and F analytic on a disk of radius $> \mathfrak{B}$ instead of $F = \exp(z \cdot)$; Corollary 12.1 gives all orders for the rows’ step set
fixed-row asymptotic (Thm. 4.1)	Theorem 3.1	weaker: $O(n^{-1/5})$ against a complete expansion; same κ_r and $c(r)$
root-product formula (5.6) (its (4.5))	Proposition 5.2	same; its Q_r is $-Q_r$ here; inside roots counted by reciprocity, as source 42 does

source 75-03	printed here	comparison
$\kappa_4 = 4(\varphi - \sqrt{\varphi})$, $\kappa_5 = \frac{5}{2}(3 - \sqrt{5})$, A_2, A_3 closed forms, constants $r \leq 8$	Section 7, Table 3	same values to every printed digit; κ_4 's closed form is already printed there and credited in OEIS to Kotesovec [2]
marked sextic (Prop. 5.1)	(4.5)	the same polynomial term by term; a third derivation (Remark 25.1)
existence of the all-orders expansion for A215570, $\alpha_{5,j} \in \mathbb{Q}(\sqrt{5})$ (Thm. 7.1)	Theorems 3.1, 3.3	same; a second route by a two-variable saddle (Section 27.1)
$\alpha_{5,1} = -13/10 + 13\sqrt{5}/50$ primitive-factor law, mean $2\kappa - 1$ (Thm. 8.1)	(3.6) Theorems 10.1, 19.2	same value; a third route, (27.3) same, plus the limiting variance (24.2)
Lambert- W_{-1} anchor, first inverse corrections (Section 9)	(11.2), (11.4), (20.3)	same; for $r = 5$ the second-order coefficient is now explicit, (28.1)
rounded threshold $\lceil x_* \mp Cx_0^{-3} \rceil$	(20.4) with $M = 2$	same; an instance of <code>p0:thm:staircase</code>
monotonicity by appending a balanced block	Section 11.3	same argument

Two increments over Parts I and II are small enough to state here.

Remark 24.1 (A rate in the one-large-summand theorem (source 75-03)). In the notation of Section 15, with $q = r$ steps of the list (15.1), source 75-03 proves for every F analytic on a disk $|z| < R$ with $R > \mathfrak{B}$

$$\frac{[\mathbf{x}^{k_N}]F(\mathcal{L}(\mathbf{p}\mathbf{x}))}{b_{k_N}} = F'(\mathfrak{B}) + O(N^{-1/5}), \quad (24.1)$$

an additive estimate (not a relative equivalence when $F'(\mathfrak{B}) = 0$). The inputs are the two bounds of Lemma 15.1 with $\varsigma = (r + 1)/2$, the tail bound $\sum_{|\mathbf{k}| > H} b_{\mathbf{k}} = O(H^{-1/2})$ (from $\mathbb{P}(S_m = 0) = O(m^{-1/2})$), and the uniform shift ratio $b_{\mathbf{k}_N - \ell}/b_{\mathbf{k}_N} = 1 + O((L + L^2)/N)$ for $L = |\ell|$ with $L^2/N \rightarrow 0$. The cutoff proof of Theorem 15.2 is run with the cutoff $H = \lfloor N^{2/5} \rfloor$ and only for $m \leq N^{1/20}$ summands, so that $(mH)^2/N \leq N^{-1/10}$; the errors are $O(H^2/N + H^{-1/2}) = O(N^{-1/5})$, and the terms $m > N^{1/20}$ are exponentially small by (15.5) and Cauchy's estimate. With $F = \exp$ this gives the leading term of Theorem 16.1 with relative error $O(N^{-1/5})$ for every finite labelled step list, and for the rows the leading term of Theorem 3.1 with relative error $O(n^{-1/5})$. Source 75-03 calls the rate “deliberately nonsharp”; for the rows' step set it is superseded by Theorem 3.1 and Corollary 12.1.

The limiting probability generating function (19.4) is v times a negative binomial law of order two, so the number J_N of primitive factors (arches) has limiting variance

$$\text{Var } J_N \longrightarrow 2\kappa(\kappa - 1), \quad (24.2)$$

in the notation of Theorem 19.2 (for the rows also $\text{Var } C_n \rightarrow 2\kappa_r(\kappa_r - 1)$, since $C_n = J_n - 1$ in Theorem 10.1); convergence of the variance follows from the convergence of all fixed moments proved there. Source 75-03 states this variance with the mean $2\kappa - 1$, which for $r = 5$ is $14 - 5\sqrt{5}$, the last column of Table 3.

24.3 Conventions of Part IV

Part IV uses the merged notation of Section 1.5. Table 2 lists every symbol of source 75-03 that is renamed; no normalization changed except the sign of Q_r . In particular source 75-03's

asymptotic coefficients d_j are the $\alpha_{5,j}$ of (3.3), its constant $\mathcal{E}_r$ is κ_r , and its excursion series Y of the sextic is the X of (4.5). Three readings are tempting and false:

- Λ_n (source 75-03's C_n , the Legendre coefficient of Theorem 25.2) is not the number C_n of interior returns of Section 10, not the functional $\mathcal{C}_N$ of (6.2), and not the lattice Λ of Section 17;
- $\mathcal{P}_n(t)$ is the Legendre polynomial with $\mathcal{P}_n(1) = 1$, not the step polynomial $P_r(u)$ of (3.1): $\mathcal{P}_5(t) \neq P_5(t)$;
- $A_5^\circ(n)$ (source 75-03's d_n) counts the zero-free excursions of (25.3); it is not an asymptotic coefficient.

The marking variables a, b, c, d, e of steps $-2, \dots, 2$ are those of (4.5); the letter a is never the sequence.

25 A positive coefficient formula for A215570

25.1 The sextic in compact form

Put

$$A = ae, \quad B = bd, \quad C = ad^2 + b^2e, \quad \bar{c} = 1 - c.$$

Then (4.5) reads

$$A^3X^6 - A^2\bar{c}X^5 + A(B - A)X^4 + (2A\bar{c} + C)X^3 + (B - A)X^2 - \bar{c}X + 1 = 0, \quad (25.1)$$

term by term the same polynomial; source 75-03 states it in this form.

Remark 25.1 (Third derivation of the sextic (source 75-03)). Let $e_1, \dots, e_4$ be the elementary symmetric functions of the roots of the kernel quartic $eu^4 + du^3 + (c - 1)u^2 + bu + a$. The six pair products u_iu_j ($i < j$) are the roots of the resolvent

$$R(v) = v^6 - e_2v^5 + (e_1e_3 - e_4)v^4 - (e_3^2 + e_1^2e_4 - 2e_2e_4)v^3 + e_4(e_1e_3 - e_4)v^2 - e_2e_4^2v + e_4^3,$$

by expansion of $\prod_{i < j} (v - u_iu_j)$. The two small roots have product $-aX$ (Lemma 4.1). Substituting $(e_1, e_2, e_3, e_4) = (-d/e, (c - 1)/e, -b/e, a/e)$ and $v = -aX$ and multiplying by e^3/a^3 gives (25.1), as a polynomial identity; its X -derivative at the origin is -1 , which selects the formal branch. This is a third route beside the quadratic factorization of source 61 and the explicit coefficient solving of source 42 (Remark 4.2); the program `code/75-03-ballot-words-verify.py` checks the resolvent and its specialization symbolically.

With $U = AX^2$, (25.1) becomes $(1 - U)^2(1 + U - \bar{c}X) + (1 + U)BX^2 + CX^3 = 0$. Since the denominators below have constant term one, it is equivalent, in formal power series, to

$$X = 1 + cX + AX^2 + BX^2 \frac{1 + AX^2}{(1 - AX^2)^2} + CX^3 \frac{1}{(1 - AX^2)^2}. \quad (25.2)$$

25.2 Zero-step deletion and Lagrange inversion

Let $A_5^\circ(n)$ count nonnegative excursions with n copies of each of the steps $-2, -1, 1, 2$ and no zero step. Removing the n zero steps of an excursion counted by $A_5(n)$ and reinserting them gives

$$A_5(n) = \binom{5n}{n} A_5^\circ(n), \quad (25.3)$$

the same deletion that gives $A_3(n)$ in Section 7. Set $c = 0$ and give A, B, C the weights $2, 2, 3$. If $\mathbf{n} = 2i + 2j + 3k$, Lagrange inversion applied to $\mathcal{Y} = zX(z^2A, z^2B, z^3C)$, which by (25.2) satisfies $\mathcal{Y} = z(1 + A\mathcal{Y}^2 + B\mathcal{Y}^2(1 + A\mathcal{Y}^2)(1 - A\mathcal{Y}^2)^{-2} + C\mathcal{Y}^3(1 - A\mathcal{Y}^2)^{-2})$ for an auxiliary variable z , gives

$$[A^i B^j C^k]X = \frac{\mathbf{n}!}{j! k! (\mathbf{n} + 1 - j - k)!} [u^i] (1 + u)^{\mathbf{n}+1-k} (1 - u)^{-2(j+k)} : \quad (25.4)$$

select j copies of the B term and k of the C term in the $(\mathbf{n} + 1)$ st power of the Lagrange formula; the remaining powers of $1 + A\mathcal{Y}^2$ have total exponent $\mathbf{n} + 1 - k$, and extracting A^i gives (25.4). Equal powers of a, b, d, e force the two monomials of C to be used equally often, say ℓ times each; then $i = n - \ell$, $j = n - 2\ell$, $k = 2\ell$, $\mathbf{n} = 4n$, and $C^{2\ell}$ contributes $\binom{2\ell}{\ell}$. Hence

$$A_5^\circ(n) = \frac{(4n)!}{(3n + 1)!} \sum_{\ell=0}^{\lfloor n/2 \rfloor} \frac{[u^{n-\ell}] (1 + u)^{4n+1-2\ell} (1 - u)^{-2n}}{(n - 2\ell)! (\ell!)^2}. \quad (25.5)$$

25.3 A Legendre coefficient

Let $\mathcal{P}_n(t)$ be the Legendre polynomial, $\mathcal{P}_n(1) = 1$. Comparing coefficients of v^n in $((1 - v)^2 - 4zv^2)^{-1/2}$ gives

$$\sum_{\ell=0}^{\lfloor n/2 \rfloor} \frac{n! z^\ell}{(n - 2\ell)! (\ell!)^2} = (1 - 4z)^{n/2} \mathcal{P}_n((1 - 4z)^{-1/2}).$$

Substituting $z = u/(1 + u)^2$ in (25.5), for which $(1 - 4z)^{1/2} = (1 - u)/(1 + u)$, proves the following formula.

Theorem 25.2 (Positive coefficient formula (source 75-03)). *For $n \geq 0$ put*

$$\Lambda_n = [u^n] (1 + u)^{3n+1} (1 - u)^{-n} \mathcal{P}_n\left(\frac{1 + u}{1 - u}\right) \quad (25.6)$$

$$= [u^n] (1 + u)^{3n+1} (1 - u)^{-2n} \sum_{k=0}^n \binom{n}{k}^2 u^k. \quad (25.7)$$

Then

$$A_5(n) = \frac{(5n)!}{(n!)^2 (3n + 1)!} \Lambda_n. \quad (25.8)$$

Proof. (25.6) and (25.8) follow from (25.5) and (25.3). The two forms of Λ_n agree because $(1 - u)^n \mathcal{P}_n((1 + u)/(1 - u)) = \sum_k \binom{n}{k}^2 u^k$, by the binomial expansion of the Legendre polynomial. $\square$

The first values are $\Lambda_0, \dots, \Lambda_5 = 1, 7, 104, 1935, 39854, 869736$, giving $A_5(0), A_5(1), A_5(2) = 1, 35, 18720$. Formula (25.7) is a finite positive coefficient computation, independent of the recurrence proved next. It agrees with the 17 independently enumerated terms of source 61, the 16 of source 42 and the 13 data terms of OEIS A215570; and at $n = 20$ and $n = 50$, where Section 13 used OEIS b-file values, its values reproduce the ratios recorded in `data/61-balanced-diagnostics.json` to all 25 recorded digits (checked at the intake, 2026-10-02).

26 Proof of Kauers–Koutschan Conjecture 15

Normalize

$$\tilde{A}_5(n) = \frac{(n!)^3((n+1)!)^2}{(5n)!} A_5(n) = \mathfrak{h}_n \Lambda_n, \quad \mathfrak{h}_n = \frac{n!((n+1)!)^2}{(3n+1)!}, \quad (26.1)$$

and put

$$\begin{aligned} \mathfrak{R}(n) &= 65n^3 + 593n^2 + 1772n + 1740, \\ \mathfrak{S}(n) &= 65n^3 + 398n^2 + 781n + 496, \\ \mathfrak{U}(n) &= 2015n^5 + 24428n^4 + 114387n^3 + 258294n^2 + 281088n + 118368, \\ \mathfrak{V}(n) &= 910n^5 + 11032n^4 + 52047n^3 + 119686n^2 + 134365n + 58980. \end{aligned}$$

Theorem 26.1 (Kauers–Koutschan Conjecture 15 (source 75-03)). *For every integer $n \geq 0$,*

$$\sum_{j=0}^3 \mathfrak{p}_j(n) \tilde{A}_5(n+j) = 0, \quad (26.2)$$

where

$$\begin{aligned} \mathfrak{p}_0(n) &= -2(n+1)(n+2)\mathfrak{R}(n), & \mathfrak{p}_1(n) &= \mathfrak{U}(n), \\ \mathfrak{p}_2(n) &= -4\mathfrak{V}(n), & \mathfrak{p}_3(n) &= 3(3n+8)(3n+10)\mathfrak{S}(n). \end{aligned} \quad (26.3)$$

Equivalently, with $K = \{1, \dots, 14\} \setminus \{5, 10\}$, the sequence satisfies the recurrence displayed in OEIS A215570 [3]:

$$\begin{aligned} &3(n+1)(n+2)^3(n+3)^4(n+4)^2(3n+8)(3n+10)\mathfrak{S}(n)A_5(n+3) \\ &- 20(n+1)(n+2)^3(n+3)^2 \prod_{k=11}^{14} (5n+k)\mathfrak{V}(n)A_5(n+2) \\ &+ 25(n+1)(n+2) \prod_{k \in K, k \geq 6} (5n+k)\mathfrak{U}(n)A_5(n+1) \\ &- 250 \prod_{k \in K} (5n+k)\mathfrak{R}(n)A_5(n) = 0. \end{aligned} \quad (26.4)$$

The form (26.4), of order three and degree 15, is the one Kauers and Koutschan guessed [7]; source 75-03 proves the normalized form (26.2) and calls its coefficients “the coefficients in Conjecture 15” without printing the unnormalized operator. The two are equivalent: with $w(n) = (n!)^3((n+1)!)^2/(5n)!$ and

$$\mathfrak{g}(n) = \frac{125 \prod_{k \in K} (5n+k)}{(n+1)(n+2)},$$

the coefficient of $A_5(n+j)$ in (26.4) is $\mathfrak{g}(n)\mathfrak{p}_j(n)w(n+j)/w(n)$ for $j = 0, 1, 2, 3$, so (26.4) is $\mathfrak{g}(n)/w(n)$ times (26.2), and $\mathfrak{g}(n) \neq 0$. This was checked symbolically at the intake (2026-10-02); the OEIS operator also annihilates the terms given by (25.8) for $0 \leq n \leq 59$.

26.1 A rational residue representation

The substitution $t = (1+u)/(1-u)$ in (25.6) gives

$$\Lambda_n = \text{Res}_{t=1} \mathcal{W}_n(t) \mathcal{P}_n(t), \quad \mathcal{W}_n(t) = \frac{2^{2n+2} t^{3n+1}}{(t-1)^{n+1}(t+1)^{n+2}}, \quad (26.5)$$

so that

$$\frac{\mathcal{W}_{n+j}}{\mathcal{W}_n} = \left(\frac{4t^3}{t^2-1} \right)^j, \quad \mathsf{L}_n(t) := \frac{\mathcal{W}'_n}{\mathcal{W}_n} = \frac{3n+1}{t} - \frac{n+1}{t-1} - \frac{n+2}{t+1}. \quad (26.6)$$

The factor $(t+1)$ in the denominator of $\mathcal{W}_n$ is essential. The coefficients

$$\begin{aligned} \mathfrak{q}_0(n) &= -18(n+1)(3n+2)(3n+4)(3n+5)(3n+7)\mathfrak{R}(n), \\ \mathfrak{q}_1(n) &= 3(n+2)(3n+5)(3n+7)\mathfrak{U}(n), \\ \mathfrak{q}_2(n) &= -4(n+2)(n+3)^2\mathfrak{V}(n), \\ \mathfrak{q}_3(n) &= (n+2)(n+3)^2(n+4)^2\mathfrak{S}(n) \end{aligned} \quad (26.7)$$

satisfy, by cancellation of factorials,

$$\mathfrak{q}_j(n) = \mathfrak{c}(n) \mathfrak{p}_j(n) \frac{\mathfrak{h}_{n+j}}{\mathfrak{h}_n}, \quad \mathfrak{c}(n) = \frac{9(3n+2)(3n+4)(3n+5)(3n+7)}{n+2}, \quad (26.8)$$

with all denominators nonzero for $n \geq 0$.

26.2 The certificate identity

Define vectors of rational functions

$$\zeta_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \zeta_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \zeta_{j+1} = \frac{(2(n+j)+1)t \zeta_j - (n+j)\zeta_{j-1}}{n+j+1} \quad (j = 1, 2).$$

By the Legendre recurrence, $\mathcal{P}_{n+j} = \zeta_{j,0}\mathcal{P}_n + \zeta_{j,1}\mathcal{P}_{n+1}$. Set

$$\begin{pmatrix} \mathcal{R}_0 \\ \mathcal{R}_1 \end{pmatrix} = \sum_{j=0}^3 \mathfrak{q}_j(n) \left(\frac{4t^3}{t^2-1} \right)^j \zeta_j. \quad (26.9)$$

The rational functions $\mathcal{U}_n(t), \mathcal{V}_n(t)$ printed in Appendix C satisfy the two identities in $\mathbb{Q}(n, t)$

$$\mathcal{R}_0 = \mathcal{U}'_n + \mathsf{L}_n \mathcal{U}_n - \frac{(n+1)t}{t^2-1} \mathcal{U}_n - \frac{n+1}{t^2-1} \mathcal{V}_n, \quad (26.10)$$

$$\mathcal{R}_1 = \mathcal{V}'_n + \mathsf{L}_n \mathcal{V}_n + \frac{n+1}{t^2-1} \mathcal{U}_n + \frac{(n+1)t}{t^2-1} \mathcal{V}_n. \quad (26.11)$$

This is an explicit finite certificate: it is verified by substituting the printed polynomials and clearing denominators. `code/75-03-ballot-words-verify.py` does exactly this with exact rational coefficients (`sympy.cancel` of both residuals over $\mathbb{Q}(n, t)$), with no numerical sampling.

Proof of Theorem 26.1. The Legendre derivative identities

$$\mathcal{P}'_n = \frac{n+1}{t^2-1}(\mathcal{P}_{n+1} - t\mathcal{P}_n), \quad \mathcal{P}'_{n+1} = \frac{n+1}{t^2-1}(t\mathcal{P}_{n+1} - \mathcal{P}_n)$$

show that (26.10)–(26.11) are precisely the coefficients of $\mathcal{P}_n$ and $\mathcal{P}_{n+1}$ in the scalar identity

$$\sum_{j=0}^3 \mathfrak{q}_j(n) \mathcal{W}_{n+j}(t) \mathcal{P}_{n+j}(t) = \frac{d}{dt} \left[\mathcal{W}_n(t) (\mathcal{U}_n(t) \mathcal{P}_n(t) + \mathcal{V}_n(t) \mathcal{P}_{n+1}(t)) \right]. \quad (26.12)$$

For each $n \geq 0$ every function here is rational in t ; the only parameter denominator of the certificate is $n+2$. The residue at a finite point of the derivative of a rational function is zero. Taking the residue at $t = 1$ and using (26.5) gives $\sum_{j=0}^3 \mathfrak{q}_j(n) \Lambda_{n+j} = 0$. Substituting (26.8) and (26.1) and cancelling $\mathfrak{c}(n)/\mathfrak{h}_n$ yields (26.2) for every $n \geq 0$. $\square$

26.3 Exact generation, and what is not asserted

The leading coefficients $\mathfrak{p}_3(n)$ and $\mathfrak{q}_3(n)$ are positive for $n \geq 0$, so the recurrence and $\Lambda_0, \Lambda_1, \Lambda_2$ determine the sequence. Working with the integers Λ_n avoids the rational normalization; (25.8) recovers $A_5(n)$. Each new term costs a bounded number of arithmetic operations; this is not a constant bit-cost claim, since the integers have linearly growing bit length.

The order is three, as conjectured. No claim of minimality over every coefficient field is made. The normalized recurrence has the exact solution $3n + 2$:

$$\sum_{j=0}^3 \mathfrak{p}_j(n)(3(n+j)+2) = 0, \quad (26.13)$$

which is why a leading asymptotic alone does not determine every formal coefficient: a second asymptotic datum is needed (Section 27).

This settles Kauers and Koutschan's Conjecture 15 [7], listed as open in Section 22. Their request for a provably correct recurrence for A215562 ($r = 4$) remains open. The route is not the creative telescoping of the sextic proposed there (Remark 4.2): zero-step deletion and Lagrange inversion reduce the diagonal to the one-variable coefficient (25.7), and the telescoping is done on its Legendre residue representation.

27 The exact coefficients $\alpha_{5,2}, \dots, \alpha_{5,8}$

27.1 A second route to the expansion, a third to $\alpha_{5,1}$

The existence of a complete expansion (3.3) for $r = 5$ with $\alpha_{5,j} \in \mathbb{Q}(\sqrt{5})$ is Theorem 3.1 with Theorem 3.3. Source 75-03 proves it again from (25.7), by a saddle point in two variables instead of three marker variables. Since $\text{CT}_v(1+v)^n(1+u/v)^n = \sum_k \binom{n}{k}^2 u^k$,

$$\Lambda_n = \text{CT}_{u,v}(1+u) \Pi(u,v)^n, \quad \Pi(u,v) = \frac{(1+u)^3(1+v)(1+u/v)}{u(1-u)^2}. \quad (27.1)$$

The logarithmic gradient of Π vanishes at $(u_0, v_0) = (\varphi^{-4}, \varphi^{-2})$, $\varphi = (1 + \sqrt{5})/2$: the v equation gives $v^2 = u$, and with $u = t^2$, $v = t$ the u equation becomes $(t+1)(t^2 - 3t + 1) = 0$, whose root in $(0, 1)$ is $t = \varphi^{-2} = (3 - \sqrt{5})/2$. There

$$\Pi(u_0, v_0) = 27, \quad 1 + u_0 = 3\varphi^{-2}, \quad H_\Pi = \begin{pmatrix} 14/15 & -1/5 \\ -1/5 & 2/5 \end{pmatrix}, \quad \det H_\Pi = \frac{1}{3},$$

H_Π being the Hessian of $\log \Pi$ in logarithmic coordinates. On the torus $|u| = u_0$, $|v| = v_0$ one has $|\Pi| \leq 27$ with equality only at (u_0, v_0) (the triangle inequalities for $1+u$ and $1+v$ already force both phases to vanish, and $|1-u| \geq 1-u_0$). The standard saddle argument (restriction to angular distance $n^{-2/5}$, a Gaussian majorant inside and an $O(27^n e^{-cn^{1/5}})$ bound outside) gives, at every finite order,

$$\Lambda_n = \frac{3\sqrt{3}\varphi^{-2}}{2\pi} \frac{27^n}{n} \left(1 + \frac{\eta_1^\Lambda}{n} + \frac{\eta_2^\Lambda}{n^2} + \dots \right), \quad \eta_j^\Lambda \in \mathbb{Q}(\sqrt{5}), \quad (27.2)$$

and with Stirling's series in (25.8) the expansion (3.3) for $r = 5$, with the constant K_5 of (3.6).

For the first correction put, for $j \geq 2$, $\ell_j^+(q) = (q\partial_q)^j \log(1+q)$ and $\ell_j^-(q) = (q\partial_q)^j [-\log(1-q)]$, and

$$\mathcal{K}_j(Z_1, Z_2) = (3\ell_j^+(u_0) + 2\ell_j^-(u_0))Z_1^j + \ell_j^+(\varphi^{-2})(Z_2^j + (Z_1 - Z_2)^j),$$

where (Z_1, Z_2) is a centred Gaussian vector with covariance $H_{\Pi}^{-1} = \begin{pmatrix} 6/5 & 3/5 \\ 3/5 & 14/5 \end{pmatrix}$. Taylor expansion of amplitude and exponent gives, with $G_0 = 1 + u_0$,

$$\eta_1^{\Lambda} = \frac{1}{G_0} \mathbb{E} \left[-\frac{u_0 Z_1^2}{2} + \frac{u_0 Z_1 \mathcal{K}_3}{6} + \frac{G_0 \mathcal{K}_4}{24} - \frac{G_0 \mathcal{K}_3^2}{72} \right] = -\frac{71}{90} + \frac{13\sqrt{5}}{50}. \quad (27.3)$$

The factorial multiplier in (25.8) has relative first correction $-23/45$, so $\alpha_{5,1} = \eta_1^{\Lambda} - 23/45 = -13/10 + 13\sqrt{5}/50$: the value of (3.6), here by a third contraction after Sections 9 and 18. The checker evaluates (27.3) exactly.

27.2 The triangular algorithm

By (3.3) and Stirling's series, $\tilde{A}_5(n)$ has a complete expansion

$$\tilde{A}_5(n) \sim n \sum_{k \geq 0} \tilde{c}_k n^{-k}, \quad \tilde{c}_0 = \varphi^{-2}, \quad \tilde{c}_1 = 1 - \frac{4\sqrt{5}}{25}, \quad (27.4)$$

where $\tilde{c}_0$ comes from K_5 and $\tilde{c}_1 = \varphi^{-2}(\eta_1^{\Lambda} + 17/9)$ from $\alpha_{5,1}$, $17/9$ being the first Stirling correction of $\mathfrak{h}_n$. Put $\hat{\mathfrak{p}}_j(z) = z^5 \mathfrak{p}_j(1/z)$. Substituting the expansion in (26.2) with $n = 1/z$ gives, for every $k \geq 2$,

$$[z^{k+2}] \sum_{j=0}^3 \hat{\mathfrak{p}}_j(z) \sum_{\ell=0}^k \tilde{c}_{\ell} z^{\ell} (1 + jz)^{1-\ell} = 0. \quad (27.5)$$

For a trial term n^{β} the first nonzero coefficient is $[z^2] \sum_j \hat{\mathfrak{p}}_j(z) (1 + jz)^{\beta} = 1625\beta(\beta - 1)$, so the two free powers are n and 1 , and the coefficient of $\tilde{c}_k$ in (27.5) is $1625k(k - 1) \neq 0$. Given $\tilde{c}_0$ and $\tilde{c}_1$, (27.5) is therefore triangular; for example $\tilde{c}_2 = 2\sqrt{5}/25$, $\tilde{c}_3 = -74\sqrt{5}/3125$, $\tilde{c}_4 = 814\sqrt{5}/78125$. Finally, as formal series in z ,

$$\sum_{k \geq 0} \alpha_{5,k} z^k = \frac{\sum_{k \geq 0} \tilde{c}_k z^k}{\varphi^{-2}(1 + z)^2} \exp \left(\sum_{j \geq 1} \frac{B_{2j}(5^{1-2j} - 5)}{2j(2j - 1)} z^{2j-1} \right), \quad (27.6)$$

with Bernoulli numbers B_{2j} : the exponential is Stirling's correction for $(5n)!/(n!)^5$, and $(1 + z)^{-2}$ that of $(n + 1)^{-2}$, since $A_5(n) = \tilde{A}_5(n)(5n)!/((n!)^5(n + 1)^2)$.

Proposition 27.1 (Exact coefficients for A215570 (source 75-03)). *For every k , the coefficient $\alpha_{5,k}$ of (3.3) is the k th coefficient computed by (27.5) and (27.6) from (27.4). The values for $k \leq 8$ are those of Table 6.*

Proof. The expansion exists (Theorem 3.1, or Section 27.1); inserting it into (26.2), which holds for every n , gives a formal identity whose coefficients are (27.5). These determine $\tilde{c}_k$ for $k \geq 2$ uniquely from $\tilde{c}_0, \tilde{c}_1$, which are fixed by K_5 and $\alpha_{5,1}$ (Theorem 3.3). Stirling's series converts $\tilde{c}_k$ into $\alpha_{5,k}$ by (27.6). The table entries are the exact values recorded by `code/75-03-ballot-words-verify.py` in `data/75-03-ballot-words-verification_results.json`, regenerated identically at the intake. $\square$

The value $\alpha_{5,2} = 0.31302\dots$ is the limit of the last column of the diagnostic table of Section 13, which source 61 did not claim as a formula: at $n = 50$ that column is 0.3111216 , and $\alpha_{5,2} + \alpha_{5,3}/50 + \alpha_{5,4}/50^2 = 0.3111216$. Table 7 shows source 75-03's numerical check of the truncations.

Table 6: The exact coefficients of (3.3) for A215570 and of (27.4) (source 75-03). The values for $k = 0, 1$ were known (Theorem 3.3); those for $2 \leq k \leq 8$ are new here. Source 75-03 prints $k \leq 4$ in its text and $k \leq 8$ in its data file. Decimals are rounded.

k	$\alpha_{5,k}$	decimal	$\tilde{c}_k$
0	1	1	$\frac{3}{2} - \frac{\sqrt{5}}{2}$
1	$-\frac{13}{10} + \frac{13\sqrt{5}}{50}$	-0.71862232585	$1 - \frac{4\sqrt{5}}{25}$
2	$\frac{36}{25} - \frac{63\sqrt{5}}{125}$	0.31302173934	$\frac{2\sqrt{5}}{25}$
3	$-\frac{407}{250} + \frac{4283\sqrt{5}}{6250}$	-0.09566733638	$-\frac{74\sqrt{5}}{3125}$
4	$\frac{29387}{15625} - \frac{64554\sqrt{5}}{78125}$	0.03312470759	$\frac{814\sqrt{5}}{78125}$
5	$-\frac{1682397}{781250} + \frac{3745709\sqrt{5}}{3906250}$	-0.00929921333	$\frac{1918\sqrt{5}}{1953125}$
6	$\frac{23801122}{9765625} - \frac{52959909\sqrt{5}}{48828125}$	0.01195322150	$\frac{248974\sqrt{5}}{48828125}$
7	$-\frac{52826879}{19531250} + \frac{118935383\sqrt{5}}{97656250}$	0.01856723270	$\frac{1667798\sqrt{5}}{244140625}$
8	$\frac{735722961}{244140625} - \frac{1619559792\sqrt{5}}{1220703125}$	0.04683285829	$\frac{21982414\sqrt{5}}{1220703125}$

Table 7: Signed relative error $(K_5 5^{5n} n^{-3} \sum_{k \leq M} \alpha_{5,k} n^{-k}) / A_5(n) - 1$ (source 75-03; rounded from its recorded results file, Section 29). Numerical illustrations, not interval-certified inequalities; the analytic argument, not this table, proves the expansion.

n	$M = 0$	$M = 1$	$M = 2$	$M = 4$	$M = 8$
20	$3.6442 \cdot 10^{-2}$	$-7.9889 \cdot 10^{-4}$	$1.2182 \cdot 10^{-5}$	$2.8011 \cdot 10^{-9}$	$-3.2337 \cdot 10^{-13}$
50	$1.4454 \cdot 10^{-2}$	$-1.2625 \cdot 10^{-4}$	$7.7105 \cdot 10^{-7}$	$2.9386 \cdot 10^{-11}$	$-7.3726 \cdot 10^{-17}$
100	$7.2066 \cdot 10^{-3}$	$-3.1432 \cdot 10^{-5}$	$9.6024 \cdot 10^{-8}$	$9.2439 \cdot 10^{-13}$	$-1.3794 \cdot 10^{-19}$
250	$2.8777 \cdot 10^{-3}$	$-5.0166 \cdot 10^{-6}$	$6.1318 \cdot 10^{-9}$	$9.5004 \cdot 10^{-15}$	$-3.5266 \cdot 10^{-23}$
1000	$7.1883 \cdot 10^{-4}$	$-3.1315 \cdot 10^{-7}$	$9.5703 \cdot 10^{-11}$	$9.2939 \cdot 10^{-18}$	$-1.3288 \cdot 10^{-28}$

Remark 27.2 (Scope (source 75-03)). Proposition 27.1 concerns the dominant exponential sector, as in the remark closing Section 6; it determines no exponentially small sector and no Stokes data. The leading characteristic polynomial of the normalized recurrence (26.2) is $65(\lambda - 1)^2(27\lambda - 2)$, here with λ a characteristic root, not $r \log r$. The double root 1 carries the two powers n and 1 above; the third root $2/27$ indicates a possible formal mode smaller by the factor $(2/27)^n$, whose contribution to the sequence is not determined here. The coefficients grow from $k = 6$ on, as expected of an asymptotic series; nothing is claimed about its summability.

28 Inversion, rounding and repository machinery

Source 75-03's inversion of A215570 is the instance $r = 5$ of Section 11 and of the rounding bound (20.4), and through them of the canonical transseries volume (Section 1.4): its Lambert anchor is (11.2), the large-solution case $b < 0$ of `p0:thm:lambert-core` (display `p0:eq:core-lambert-minus`, with $a = \lambda = 5 \log 5$ and $b = -\beta = -3$); its corrections are the first two coefficients of the Lambert-centred reversion `p0:thm:lambert-centered`, which is the explicit inverse `t2:thm:balanced-inverse` (display `t2:eq:balanced-first`) of the companion volume `Analysis/Transseries/docs/series-and-transseries/Combinatorial_Transseries_Inverses/Combinatorial_Transseries_Inverses.tex`; and its rounded threshold is parts (2)–(3) of `p0:thm:staircase` (Remark 20.1). Source 75-03 cites none of these, and no novelty is claimed for them. What is new is one value: with $\alpha_{5,2}$ known, (11.4) for $r = 5$ is explicit to second order,

$$n = x_0 + \frac{13(5 - \sqrt{5})}{50\lambda} \frac{1}{x_0} + \left(\frac{83\sqrt{5} - 213}{500\lambda} + \frac{39(5 - \sqrt{5})}{50\lambda^2} \right) \frac{1}{x_0^2} + O(x_0^{-3}), \quad (28.1)$$

for $Y = A_5(n)$, $x_0 = x_0(Y)$, $\lambda = 5 \log 5$; here $b_2 = \alpha_{5,2} - \frac{1}{2}\alpha_{5,1}^2 = (213 - 83\sqrt{5})/500$, and the two coefficients are $\mu_1 = 0.0893010312$ and $\mu_2 = 0.0264800992$. The second term accounts for the residual column of the inverse diagnostic table in Section 13: μ_2/x_0^2 is $4.150 \cdot 10^{-4}$, $1.035 \cdot 10^{-4}$ and $1.059 \cdot 10^{-5}$ at $n = 8, 16, 50$, against the printed $-4.164 \cdot 10^{-4}$, $-1.038 \cdot 10^{-4}$, $-1.060 \cdot 10^{-5}$ (the column is $-\mu_2/x_0^2 + O(x_0^{-3})$). Source 75-03 states the rounded threshold

$$[x_*(Y) - Cx_0^{-3}] \leq N_5(Y) \leq [x_*(Y) + Cx_0^{-3}],$$

with x_* the truncation of (28.1) after x_0^{-2} ; this is (20.4) with $M = 2$, since x_0^{-3} and $(\log Y)^{-3}$ are comparable. It also carries the reversion of the anchor one order beyond (20.3): with $D_Y = \log(Y/K_5) - 3 \log \lambda$, the number $u = \lambda x_0$ solves $u - 3 \log u = D_Y$, and

$$\lambda x_0 = D_Y + 3 \log D_Y + \frac{9 \log D_Y}{D_Y} + \frac{27 \log D_Y - \frac{27}{2}(\log D_Y)^2}{D_Y^2} + O\left(\frac{(\log D_Y)^3}{D_Y^3}\right), \quad (28.2)$$

the nested-logarithm expansion that follows `t2:thm:balanced-inverse` in the companion volume. Source 75-03 adds that for other rows the general theorem “currently supplies only the coarser $O(x^{-1/5})$ uncertainty”; in this report Theorem 3.1 gives every row a complete expansion, so Section 11 applies to every row at every order.

29 Source 75-03's computations

The shipped `code/75-03-ballot-words-verify.py`, with its integer certificate arrays in `code/75-03-ballot-words-certificate_data.py`, performs these exact checks:

- (a) the generic quartic pair-product resolvent of Remark 25.1 and its specialization to the sextic, as polynomial identities;
- (b) both certificate identities (26.10)–(26.11) identically in $\mathbb{Q}(n, t)$, the four relations (26.8), and the solution (26.13);
- (c) the Hessian H_Π and the Gaussian expectation (27.3) in $\mathbb{Q}(\sqrt{5})$;
- (d) agreement of the recurrence-generated Λ_n with (25.7) for $0 \leq n \leq 103$;
- (e) an independent content-state dynamic program, the recursion of Section 13 (no row recurrence is used), for $r = 2, 3, 4$ ($n \leq 10$), $r = 5$ ($n \leq 8$), $r = 6$ ($n \leq 5$) and $r = 7$ ($n \leq 4$), and the thirteen OEIS data terms of A215570.

It also generates Λ_n for $n \leq 1000$ by the recurrence, writes Λ_n and $A_5(n)$ for $n \leq 103$ to `data/75-03-ballot-words-a215570_terms.csv`, and records $\alpha_{5,0}, \dots, \alpha_{5,8}$, $\tilde{c}_0, \dots, \tilde{c}_8$, the constants of Table 3 for $r \leq 8$ (to 35 digits), the dynamic-programming rows and Table 7 in `data/75-03-ballot-words-verification_results.json`; its standard output is `data/75-03-ballot-words-verification_log.txt`. The recorded run used Python 3.13.5, SymPy 1.14.0 and mpmath 1.3.0 and took 3.6 s. The 53 dynamic-programming terms equal source 61’s enumerations (`data/61-balanced-exact_rows.json`). The intake reran the program on a copy (Python 3.14.4): all six checks passed, the CSV was regenerated byte for byte, and the JSON differs only in the recorded Python version and running time (and in its line endings: on Windows the program writes it with CRLF, while the shipped file has LF).

Source 75-03 marks the boundaries of these checks. The certificate check is finite symbolic algebra: it does not replace the derivation from words to (25.7), which is the proof above, and “finite agreement alone would not prove the conjecture”. Numerical errors are high-precision illustrations, not interval enclosures. “No Lean files are supplied as if they had been compiled.”

A A reproducible all-order construction

For clarity, the finite procedure implicit in the proof can be organized as follows. The input is a fixed r and a requested relative order M .

Critical algebraic data. Construct Q_r , remove $(u-1)^2$, and isolate the strict inside cluster. Represent its symmetric functions in an algebraic number field. Solve (6.3) recursively for a finite Taylor jet of $\tau(\mathbf{y})$; obtain ρ , Φ , and the tilted probabilities. Internal root collisions can be handled by the coefficients of the cluster factor instead of individual root labels.

Puiseux data. Set $t = \rho(\mathbf{y})(1 - w^2)$. Substitute $u_0 = \tau(\mathbf{y})(1 + \sum_{j \geq 1} a_j(\mathbf{y})w^j)$ in the kernel, choosing $a_1 = -\sqrt{2/V(\mathbf{y})}$. The remaining coefficients are obtained recursively. Multiply by the analytic noncritical cluster product and the factor $1/t$ to form the needed Puiseux jet of $\mathcal{E}$ and, independently, of $\log \mathcal{E}$.

Length coefficients. For every odd power w^{2j+1} , use (6.11) and the gamma-ratio expansion. The even powers belong to the local analytic part. Retain enough terms for the requested order, and include the length-period factor ν .

Composition coefficients. Expand the phase about zero, substitute $y = ix/\sqrt{N}$, and multiply the amplitude jets. Replace each even monomial by its centered Gaussian moment with covariance H^{-1} ; discard odd monomials. Wick's rule evaluates these moments as finite sums over pairings. The determinant and common π factor are kept outside the normalized series.

Calibration and inversion. Divide the resulting excursion expansion by the logarithmic reference expansion if desired. The exact coefficient (5.3) and Stirling's series are an independent normalization check. Convert $N = rn$, compute the formal logarithm of the unit series, and use (11.3) for inverse coefficients.

Every stage is finite at fixed r, M . The procedure proves computability, not an optimal complexity bound or a claim that a complete general-purpose implementation is included in this package. The supplied exact symbolic implementation covers the first correction for $r = 5$. Source 42's formulation of the same procedure in lattice coordinates is (17.10)–(17.12), its first-order specialization is (18.2), and its program covers the first correction for $r = 3$ and $r = 5$. For $r = 5$, source 75-03's program (Part IV) replaces the saddle construction above the first order by the recurrence algorithm (27.5)–(27.6) and covers every order; it has recorded $\alpha_{5,j}$ for $j \leq 8$.

B Package contents and build instructions

The report ships the programs and recorded outputs of both sources under prefixed names. Source 61 contributes `code/61-balanced-verify.py` and `code/61-balanced-derive_alpha5.py`, their outputs `data/61-balanced-*`, its source and claim provenance notes `61-balanced-SOURCES.md` and its build receipt `61-balanced-BUILD.md`. Source 42 contributes `code/42-late-growing-verify.py` and `code/42-late-growing-first_correction.py` and their outputs `data/42-late-growing-*`. The programs read and write their outputs under the delivered, unprefixed names in a `data` directory beside `code`, and both verification programs write a file named `verification.txt`; they must therefore be run on separate copies in the delivered layout. The report's README gives the commands.

Source 75-03 (Part IV, added October 2, 2026) contributes `code/75-03-ballot-words-verify.py` and its certificate arrays `code/75-03-ballot-words-certificate_data.py`, the recorded standard output `data/75-03-ballot-words-verification_log.txt`, the exact results `data/75-03-ballot-words-verification_results.json`, and Λ_n and $A_5(n)$ for $n \leq 103$ in `data/75-03-ballot-words-a215570_terms.csv`. Its program imports `certificate_data` by that delivered name and writes its JSON and CSV beside itself under the delivered names; it must likewise be run on a copy under those names, never in this directory.

The computations use arbitrary-precision integers, rational arithmetic, SymPy and mpmath (the shipped `data/61-balanced-requirements.txt`, whose content is identical to source 42's requirements file); they need no network access. Source 42's `-quick` switch reduces its dynamic-programming boxes; it changes the recorded finite ranges, not the mathematical results, and it overwrites the diagnostic data with the smaller run. Source URLs and the origin of the two externally supplied large-index values are recorded in `61-balanced-SOURCES.md`. No external paper is redistributed. The report is built with pdfL^AT_EX; its bibliography is embedded, so no BibTeX run is needed.

C The full rational telescoping certificate (source 75-03)

This appendix belongs to Part IV and gives every coefficient needed in (26.10)–(26.11). Put $m = n + 2$ (a local letter, not a length), and

$$\mathfrak{R}_\#(m) = 65m^3 + 203m^2 + 180m + 48, \quad \mathfrak{S}_\#(m) = 65m^3 + 8m^2 - 31m + 6,$$

which are $\mathfrak{R}(n)$ and $\mathfrak{S}(n)$ of Section 26 after the shift. Let

$$\mathcal{U}_n(t) = -\frac{2t}{m(t^2 - 1)^2} \sum_{j=0}^8 \mathfrak{u}_j(m)t^j, \quad \mathcal{V}_n(t) = \frac{2t^2}{m(t^2 - 1)^2} \sum_{j=0}^8 \mathfrak{v}_j(m)t^j. \quad (\text{C.1})$$

The coefficient polynomials $\mathfrak{u}_j, \mathfrak{v}_j$ (source 75-03's α_j, β_j , not the asymptotic coefficients) are printed below, split into powers of t so that no numerator term is omitted. The only parameter denominator is $m = n + 2$, harmless for $n \geq 0$.

C.1 Coefficients of $\mathcal{U}_n$

$$\begin{aligned} \mathfrak{u}_0 &= 9m(m-1)(3m-2)(3m-1)(3m+1)\mathfrak{R}_\#(m), \\ \mathfrak{u}_1 &= 3m(3m-2)(3m-1)(3m+1)\mathfrak{R}_\#(m), \\ \mathfrak{u}_2 &= -3m(3m-1)(3m+1)(28m^2 - 47m + 18)\mathfrak{R}_\#(m), \\ \mathfrak{u}_3 &= -6m(3m+1)(15m^2 - 18m + 5)\mathfrak{R}_\#(m), \\ \mathfrak{u}_4 &= (3m+1)(13195m^7 + 14169m^6 - 32535m^5 - 19461m^4 \\ &\quad + 15480m^3 + 5472m^2 - 1888m - 384), \\ \mathfrak{u}_5 &= 11765m^7 + 28943m^6 + 2891m^5 - 27207m^4 \\ &\quad - 12320m^3 + 5368m^2 + 2848m - 384, \\ \mathfrak{u}_6 &= -10920m^8 - 23899m^7 + 10511m^6 + 42923m^5 \\ &\quad + 12369m^4 - 16208m^3 - 5624m^2 + 2368m + 384, \\ \mathfrak{u}_7 &= -16(m-1)(m+1)(m+2)^2\mathfrak{S}_\#(m), \\ \mathfrak{u}_8 &= 32m(m-1)(m+1)(m+2)^2\mathfrak{S}_\#(m). \end{aligned}$$

C.2 Coefficients of $\mathcal{V}_n$

$$\begin{aligned}
v_0 &= -3m(m-1)(3m-2)(3m-1)(3m+1)\mathfrak{R}_\#(m), \\
v_1 &= -6m(m-1)(3m-1)(3m+1)\mathfrak{R}_\#(m), \\
v_2 &= -6m(3m+1)(1300m^6 + 3475m^5 + 1269m^4 \\
&\quad - 1895m^3 - 841m^2 + 244m + 48), \\
v_3 &= -2m(3m+1)(715m^5 + 2103m^4 + 1265m^3 \\
&\quad - 523m^2 - 168m + 208), \\
v_4 &= 43745m^8 + 126154m^7 + 67228m^6 - 57642m^5 \\
&\quad - 46597m^4 + 5216m^3 + 6760m^2 - 480m - 384, \\
v_5 &= 24(m+1)^2(325m^5 + 495m^4 - 330m^3 \\
&\quad - 202m^2 + 148m - 16), \\
v_6 &= -16(m+1)(1430m^7 + 3556m^6 + 471m^5 \\
&\quad - 2714m^4 - 307m^3 + 684m^2 - 72m - 24), \\
v_7 &= -16(m+1)(m+2)^2(2m-1)\mathfrak{S}_\#(m), \\
v_8 &= 32m(m+1)(m+2)^2(2m-1)\mathfrak{S}_\#(m).
\end{aligned}$$

C.3 How to verify the certificate without a guesser

Form $\mathcal{R}_0, \mathcal{R}_1$ from the vectors ζ_j of Section 26, insert (C.1) into the right sides of (26.10)–(26.11), subtract the left sides and multiply by common denominators in $\mathbb{Q}[n, t]$: every resulting polynomial coefficient is zero. The calculation involves no root choices, no limits and no sequence values. The shipped `code/75-03-ballot-words-certificate_data.py` stores the same u_j, v_j as integer coefficient arrays (in descending powers of m), and `code/75-03-ballot-words-verify.py` reduces the two residuals exactly to zero; the printed polynomials were expanded and compared with those arrays at the intake (2026-10-02), with no difference. The recurrence then needs only the residue representation (26.5) and the vanishing residue of a rational derivative.

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